Curs 1 - Introducere

A system can be any collection of hardware, software, and firmware components.

Metrics = criteria used to evaluate the performance of a system
Workloads = the requests maade by the users of a systems

Why do we need EP?

- Determine a system's capabilities
- Determine feasibility of future expansion
- System comparison
- Identify performance problems/bottlenecks
- System tuning

Process

1. Select appropriate evaluation techniques, performance metrics, system workload
2. Conduct performance measurements
3. Use statistical techniques for comparing alternatives
4. Design simulation/measurement experiments to get maximum results with least effort
5. Perform simulations correctly
6. Mathematical techniques

Main parts

1. Overview of Performance Evaluation
2. Measurement Techniques and Tools
3. Probability Theory and Statistics
4. Experimental Design and Analysis
5. Simulation
6. Queueing Theory

Measurement techniques and tools

- Types of workloads
- Popular benchmarks
- The art of workload selection
- Workload characterization techniques
- Monitors
- Accounting logs
- Monitoring distributed systems
- Load drivers
- Capacity planning
- The art of data presentation
- Ratio games

Probability theory and statistics

- Probability and statistics concepts
- Summarizing measured data by a single number
- Summarizing the variability of measured data
- Graphical methods to determine distributions of measured data
- Sample statistics
- Confidence interval
- Comparing two alternatives
- Measure of relationship
- Simple and multiple linear regression models
- Other regression models

Experimental. design and analysis

- Introduction to experimental design
- 2^k factorial designs
- $2^k r$ factorial designs with replications
- 2^{k-P} fractional factorial designs
- One factor experiments
- Two factors full factorial design without replications
- Two factors full factorial design with replications
- General full factorial designs with *k* factors

Simulation

- Introduction to simulation
- Types of simulation
- Model verification and validation
- Analysis of simulation results
- Random-number generation
- Testing random-number generators
- Random-variate generation
- Commonly used distributions

Example

- What performance metrics should be used to compare the performance of two disk drives?
- Which type of monitor (software or hardware) would be more suitable for measuring each of the following quantities:
  - Number of Instructions executed by a processor?

- Response time of packets on a network?

Example (cont)
The number of packets losts on two links was measured for four files sizes as shown below:

File size	Link A	Link B
50	0	1
1000	5	10
1400	7	3
1800	3	0

Which link is better?

Example (cont)

The performance of a system depends on the following three factors:

- Garbage collection technique used: Gl, G2, or none.
- Type of workload: editing, computing, or Al.
- Type of CPU: Cl, C2, or C3.

How many experiments are needed?

How does one estimate the performance impact of each factor?

Examples (cont)

In order to compare the performance of two cache replacement algorithms:

- What type of simulation model should be used?
- How long should the simulation be run?
- What can be done to get the same accuracy with a shorter run?
- How you decide if the random-number generator in the simulation is a good generator?

The art of performance evaluation

Example:

• These are the throughputs of two systems A and B in transactions per second

System	Workload 1	Workload 2
A	20	10
B	10	20

Possible solutions

1. Compare the average

System	Workload 1	Workload 2	Average
A	20	10	15
B	10	20	15

Conclusion: The two systems are equally good

2. Compare the ratio with system B as the base

System	Workload 1	Workload 2	Average
A	2	0.5	1.25
B	1	1	1

Conclusion: System A is better than B

3. Compare the ratio with system A as the base

System	Workload 1	Workload 2	Average
A	1	1	1
B	0.5	2	1.25

Conclusion: System A is better than B

4. Similar games in: Selection of workload, Measuring the systems, Presenting the results

EP curs 2 – Common mistakes and metrics

Common mistakes

1. No goals
2. Biased goals
3. Unsystematic approach
4. Analysis without understanding the problem
5. Incorrect performance metrics
6. Unrepresentative workload
7. Wrong evaluation technique
8. Overlook important parameters
9. Ignore significant factors
10. Inappropriate experimental design
11. Inappropriate level of detai
12. No analysis
13. Erroneous analysis
14. No sensitivity analysis
15. Ignoring errors in input
16. Improper treatment of outliers
17. Assuming no change in the future
18. Ignoring variability
19. Too complex analysis
20. Improper presentation of results
21. Ignoring social aspects
22. Omitting assumptions and limitation

A systematic approach to pert evaluation

1. State goals and define the system
2. List services and outcomes
3. Select metrics
4. List parameters
5. Select factors to study
6. Select evaluation technique
7. Select workload
8. Design experiments
9. Analyze and interpret data
10. Present results

    Repeat

Example

1. System definition
   The network connecting two hosts.
2. Services
   Instant messaging application (including file transfers).
3. Metrics
   - No errors or failures.
   - Rate (throughput), time (latency, rtt), resource per service.

   Leads to these metrics:
   i. Time per message or file
   ii. Maximum transfer rate per unit of time
   iii. Local and remote CPU time per transmission
4. System parameters
   - Speed of local CPU, remote CPU, network
   - Operating system
   - Network reliability
5. Workload parameters
   - Time between successive message/file transmissions
   - Sizes of messages/files
   - Type of channel
   - Other loads on the local and remote CPUs and the network
6. Factors
   - Type of channel
   - Size of network
   - Size of message/file transmitted
   - Keep fixed: type of CPU, OS, no other loads on hosts and network
7. Evaluation technique
   - Implement prototype => measurements
   - Use analytical modeling for validation
8. Workload
   - Synthetic program generating the specified types of channel requests
   - Null channel requests => resources used in monitoring and logging
9. Experimental design
   - A full factorial experiment design e.g. 2*3*3
10. Data Analysis
11. Data Presentation

Summary

- Analysis techniques, metrics, workload - depend on the goal
- Metrics are based on the services provided by the system
- System and workload parameters determine the right set of experiments

Metrics

Selecting evaluating techniques

Selecting metrics

Techniques

- Mathematical Modeling
- Simulation
- Measurement

Mathematical Modeling

- Skill set required
  - Math
- Abstraction of system
  - High-level
- Assumptions
  - Many
- Results
  - Asymptotic results, low accuracy
- Outliers
  - Not captured

Simulation

- Represents an almost "real-world" view of the system
- System can be modeled as a set of interacting subsystems and objects
- Various levels of abstraction possible
- Fine-grained performance metrics can be measured
- Coding errors
- Does not capture real-world environment

Measurement

- Testbed
- Experimental results

Selecting and evaluation technique

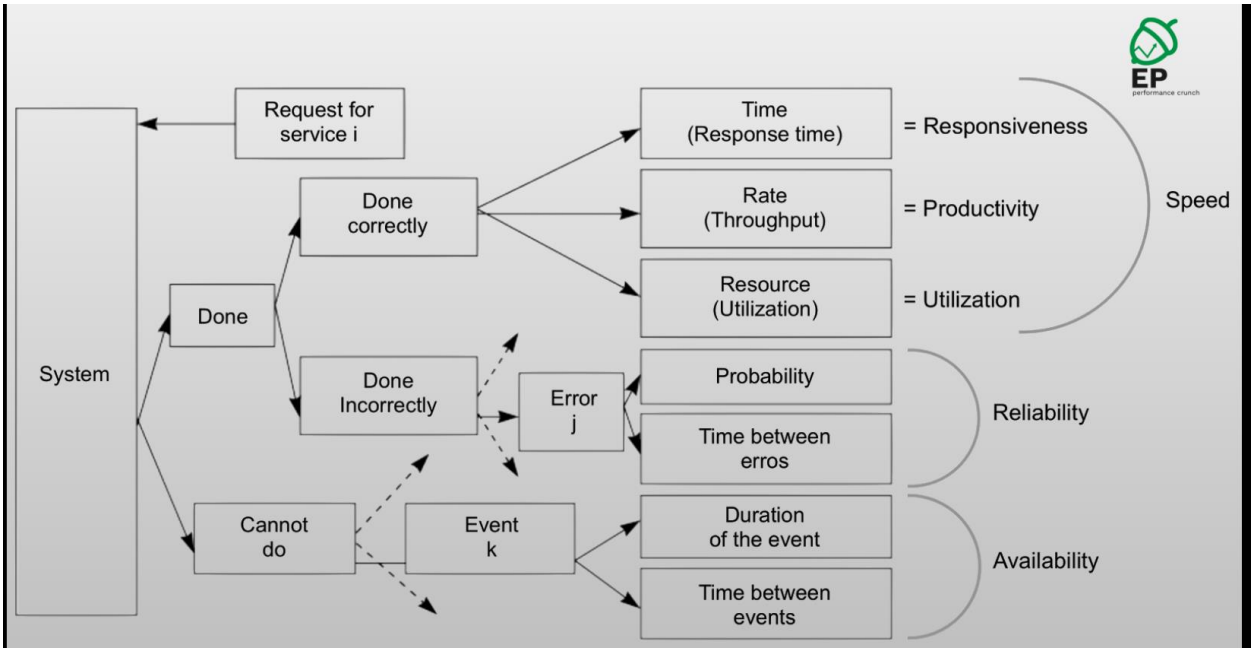
Criterion	Analytical Model	Simulation	Measuring
1. Stage	Any	Any	Prototype
2. Time required	Small	Medium	Varies
3. Tools	Analysts	Computer languages	Instrumentation
4. Accuracy	Low	Moderate	Varies
5. Trade-off evaluation	Easy	Moderate	Difficult
6. Cost	Low	Medium	High
7. Saleability	Low	Medium	High

Rules of validation

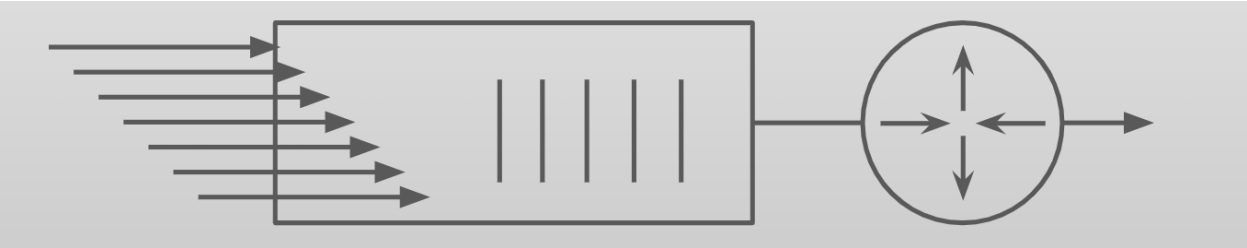
- Do not trust the results of **a simulation model** until validated with analytical modeling or measurements
- Do not trust the results of **an analytical model** until validated with a simulation model or measurements
- Do not trust the results of **a measurement** until validated with simulation or analytical modeling

Performance metrics

- List services and outcomes

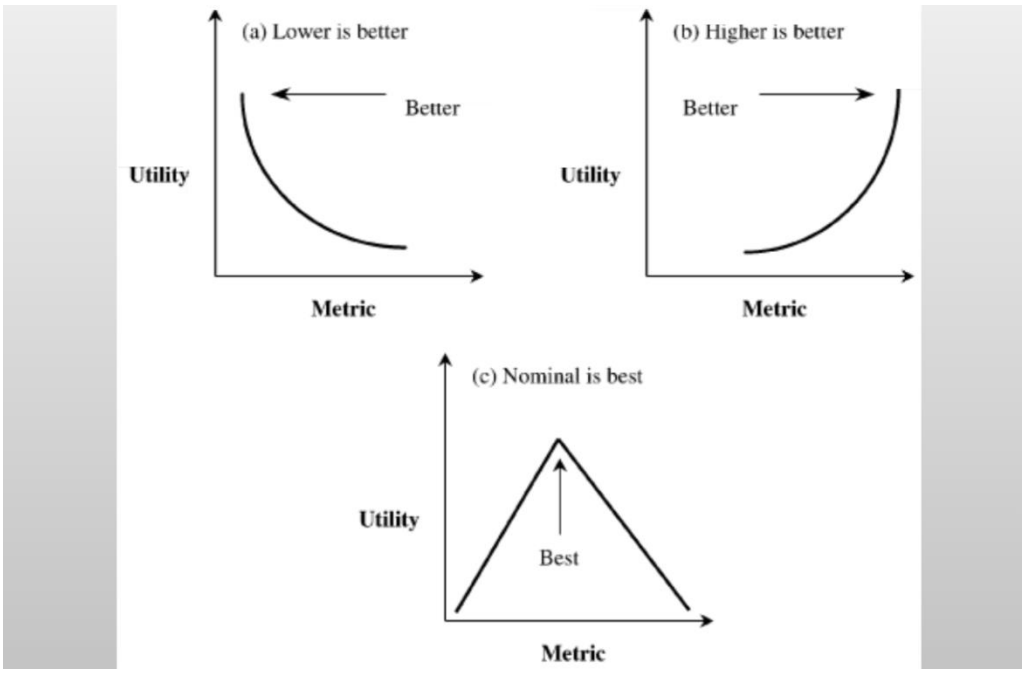


Global vs. individual



Common metrics

- Response time
  - Interval between a user's request and the system response
- Turnaround time
  - For batch stream
  - Time between the submission of a batch job and the completion of its output
- Throughput
  - The rate (requests per unit of time) at which the requests can be serviced by the system
- Nominal capacity
  - The maximum achievable throughput under ideal workload conditions
  - Bandwidth
- Usable capacity
  - The maximum throughput achievable without exceeding a prespecified response time limit (Often the response time at maximum throughput is too high to be acceptable.)
- Efficiency
  - Usable capacity / nominal capacity
- Utilization
  - The fraction of time the resource is busy servicing requests
- Reliability
  - The probability of errors or the mean time between errors
- Availability
  - The fraction of the time the system is available to service users' requests
  - Mean Time To Failure (MTTF) - mean uptime
- Cost/performance ratio
  - Metric for comparing systems
  - E.g. two transaction processing systems may be compared in terms of dollars per TPS



Requirements

- "The system should be both processing and memory efficient. It should not create excessive overhead."
- "There should be an extremely low probability that the network will duplicate a packet, deliver a packet to the wrong destination, or change the data in a packet."

- Unacceptable
  - Non-Specific
  - Non-Measurable
  - Non-Acceptable
  - Non-Realizable
  - Non-Thorough

Curs 3 – Workloads

Lecture objectives

1. Types of workloads
2. Workloads selection

Test workload types

- Real workload
    - System being used for normal operations (by actual people)
- Synthetic workload
    - Similar to real workload
    - Can be applied repeatedly in a controlled manner
    - No large real-world data files
    - No sensitive data
    - Easily modified without affecting operations
    - Easily ported to different systems due to its small size
    - May have built-in measurement capabilities

Desired characteristics of a workbench

- Representative
    - of an application area
- Portable
    - run on many different architectures and implementations
- Unbiased
    - not designed to favour a particular system
- Scalable
    - run on both small and large systems
- Measurable
    - easy to measure => broad acceptance
- Repeatable
    - minimum variance
- Explainable
    - single number easy to understand

Workloads for computer systems

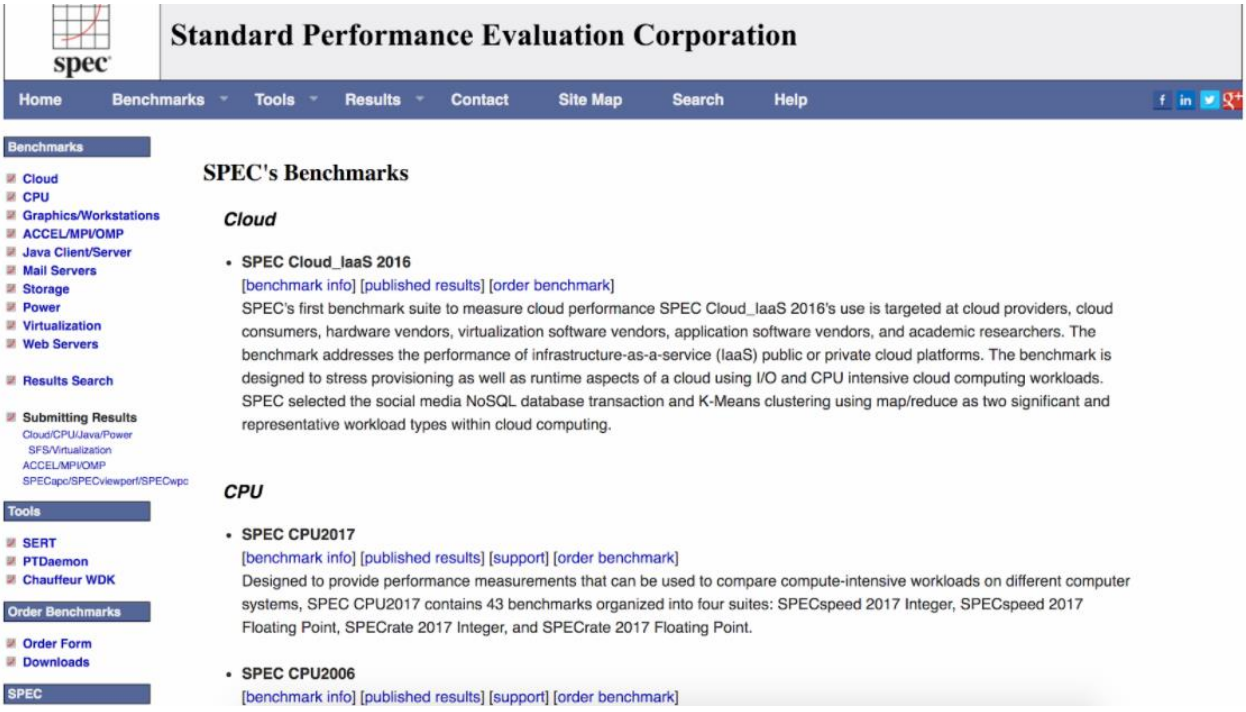
1. Add instructions
2. Instruction mixes - usage frequency
3. Kernels - matches order - micro-benchmark

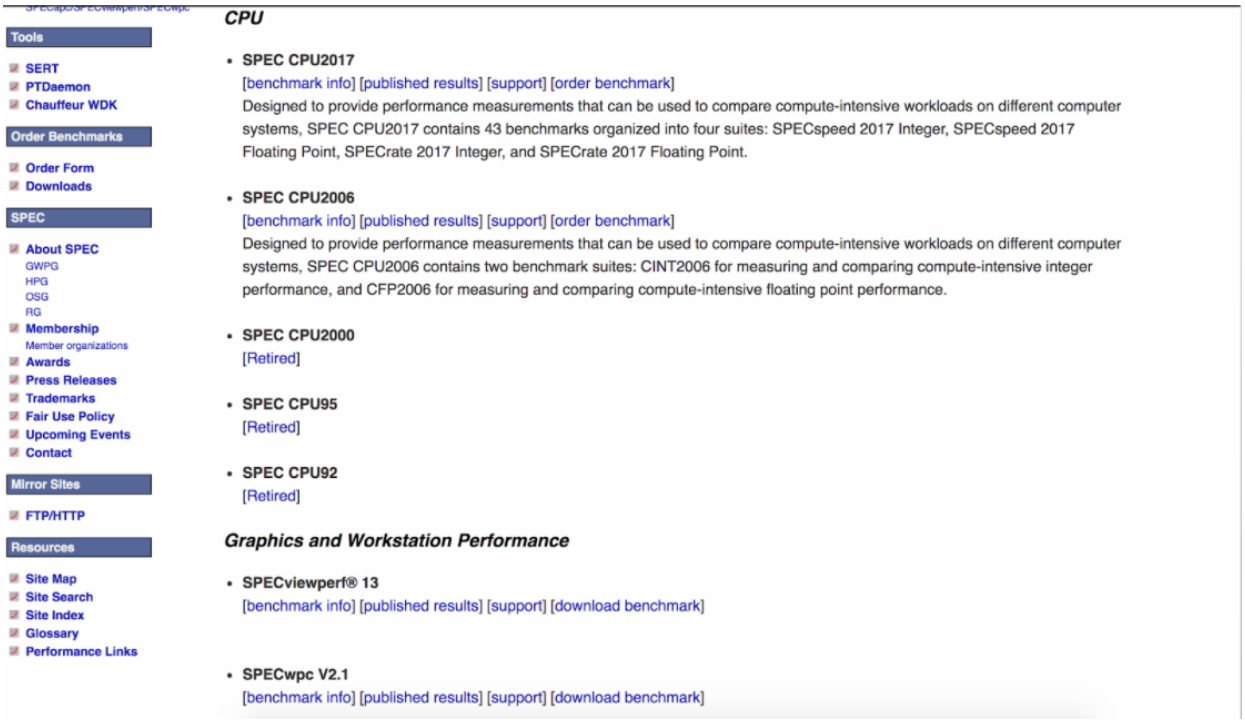
4. Synthetic programs - bigger programs
5. Application benchmarks - matches I/O

Synthetic programs

- Synthetic programs
    - Whetstone (1972) - floating point intensive numerical code
    - Linpack (1976) - linear algebra package - still used - top5000.org to rank supercomputers
    - Dhrystone (1984) - integer only, system code
- Vendor specific benchmarks
    - WinSAT (Windows System Assessment Tool)
        - Control Panel —> Performance Information and Tools
        - Reports Windows Experience Index (WEI)
        - Five sub-scores: processor, memory, 2D graphics, 3D graphics, disk
        - Overall = min of five sub-scores o SAP, Lotus/IBM, Oracle, etc
    - Use binary / machine code
- Industry benchmarks associations
    - SPEC (Systems Performance Evaluation Cooperative)
    - BAPCo (Business Applications Performance Corporation)
    - EEMBC (The Embedded Microprocessor Benchmark Consortium)
    - TPC (Transaction Performance Council) - database systems
    - SPC (Storage Performance Council) - storage systems
- https://en.wikioedia.org/wiki/Benchmark (computing)

SPEC Benchmarks





SPEC CPU benchmark

- Consists of many programs
- Ported to various platforms to remove dependency on I/O or Oss
- Compilation and execution is automated
    - Teste supplies a configuration file with location of C compiler, compilation flags, etc
- Speed
    - Execution time on the system normalised by a well known system
- Geometric mean of ratios is reported as SPECint or SPECfp
- Rate - throughput - jobs/s. Geometric mean = SPECrate

Benchmark challenges / issues

Main problem - optimising/tuning to score higher on benchmarks

- Optimised compilations not representative of real-world
    - Baseline and peak numbers (vendors publicise only peak numbers)
- n-CPU systems run n copies in parallel
    - Not representative since there is no queue of waiting jobs
- Source codes of commonly used programs are not available for use in benchmarks
    - Mostly open source
- Long run times allow jobs to run from cache
    - Not representative

- High performance on high priced systems
    - Price/performance ratio

Summary

- Benchmarks have moved from instructions (MIPS), kernels, synthetic benchmarks to industry standard benchmarks.
- Benchmarks should be representative, portable, unbiased, scalable, measurable, repeatable and explainable.
- Difficult to ensure representativeness of benchmark (compiler flags, run time, configurations - can be manipulated)
- Price and energy consumption are important along with performance

Workload Selection

Workload Selection

- Services exercised
- Level of details
- Loading level
- Timeliness

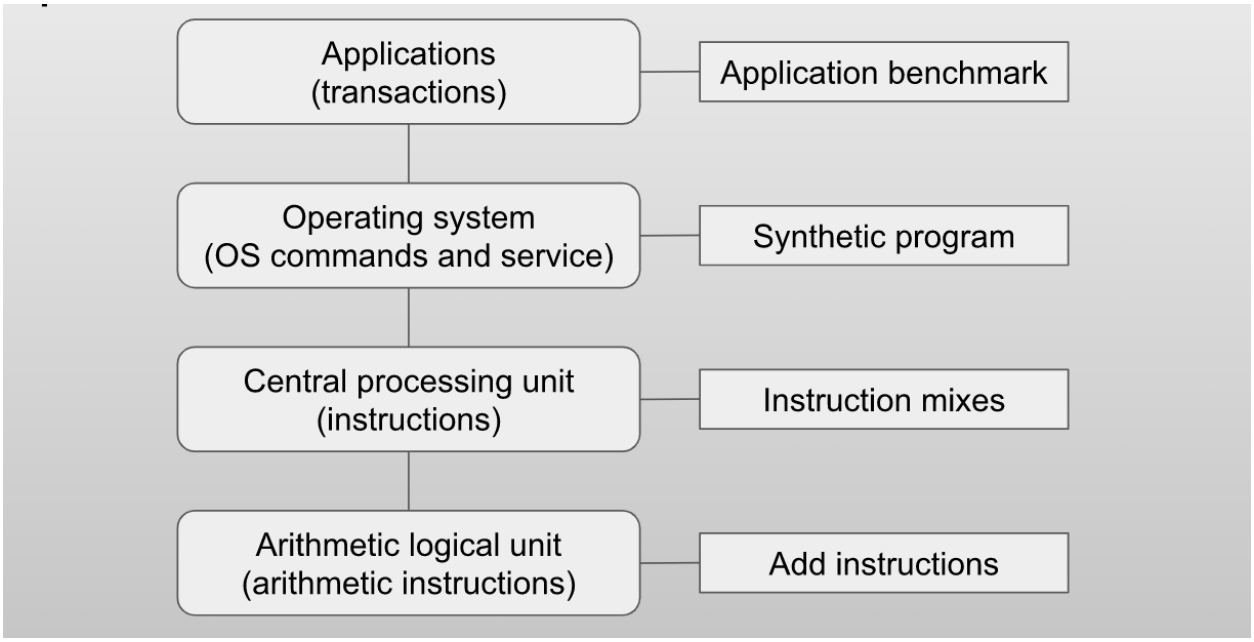
Services exercised

System services determine the workload and metrics

System <- Services <- Metrics and Workload

- System Under Test (SUT) vs Component Under Study (CUT)
- Metrics depend upon SUT
    - MIPS is ok for 2 CPUs, but not for 2 time-sharing systems
- Workload depends upon SUT
- Examples
    - CPU – instructions
    - System - transactions
    - Multiple services - exercise as complete a set of services as possible

Example



Level of detail

- Most frequent operations
  - Only if one service is significantly more frequent than others
  - Example: Add instructions
- Frequency of request types
  - Example: Instruction mixes
- Time-stamped sequence of requests
  - Probably too detailed
  - May require exact reproduction of component behaviour
- Average resource demand
- Distribution of resource demands

Representativeness

- The test workload and the real workload should have the same:
  - Elapsed time
  - Resource demands
  - Resource usage profile - sequence and amounts in which the different resources are used

Timeliness

It has to represent today's usage

Other considerations

- Loading level - a workload may exercise a system to its:
    - Full capacity (best case)
    - Beyond its capacity (worst case)
    - At a load level observed in real world (typical case)
- Impact of external components
    - Do not use a workload that makes external component a bottleneck => all alternatives to this system give equally good performance.
- Repeatability

Summary

- Services determine the workload (and metrics)
- Level of detail of the workload should match that of the model being used
- Workload should be representative of the real systems' usage in recent past

Curs 4 – Charts

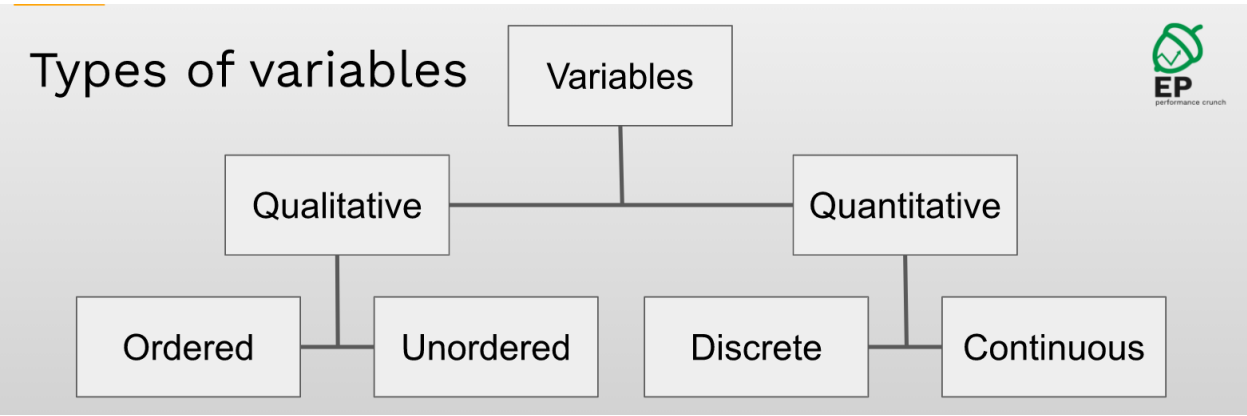
The aim of performance analysis

1. What is the aim of every performance analysis?
   Help in making a decision
2. **Presentation of final results** - one of the important steps in every performance evaluation study

Why use graphs

1. Save readers' time
   a. Present the same information more concisely
2. Interest the reader
   a. Easier to look at the figures to quickly grasp the main points of the study and read the text only for details
3. Emphasize or clarify a point
4. Reinforce a conclusion
5. Summarize the results of a study

Types of variables



- **Ordered** - Computer type: supercomputers, minicomputers, microcomputers
- **Unordered** - Workload type: scientific, engineering, educational
- **Discrete** (Integer) - Number of processors in a multiprocessor system. Size of disks in blocks.
- **Continuous** (Real) - System response time for a job.

The type of chart to be used depends on the type of variable

1. A **line** chart is used to show the ***relationship*** between two continuous **variables**.

2. A **column** chart or bar chart is used when the ***independent*** variable (also called x-variable) is a **discrete** or a **qualitative** variable.

Common chart types

- Line chart
    - Usually shows change over time
    - Compare things over time
- Bar chart
    - Compare similar things in a single category
- Double bar chart
    - Compare data
- Histogram
    - Type of bar chart
    - Compares ranges of data in a single category
- Stennplots
    - Like histograms, except they show each data point
- Pie chart
    - Shows parts of a whole
    - Often shows percentages
- Scatter plot
    - Dots of information that are not connected
    - Often show a correlation
- Timeplot
    - Shows how a variable changes over time
    - Convention
        - time on OX axis
        - values of the variable on the OY axis
- Box and whisker
    - Show variation among your data

Guidelines for good graphs

1. Require minimum effort from the reader
    a. Legends vs direct labeling
2. Maximise information - self contained
    a. CPU usage vs Daily CPU usage
3. Minimise the amount printed
    a. Grid lines - shown only if necessary to read accurate values
4. Use commonly accepted practices
    a. What people expect: origin in (0,0), linear scales, increase left to right, etc.
5. Avoid ambiguity

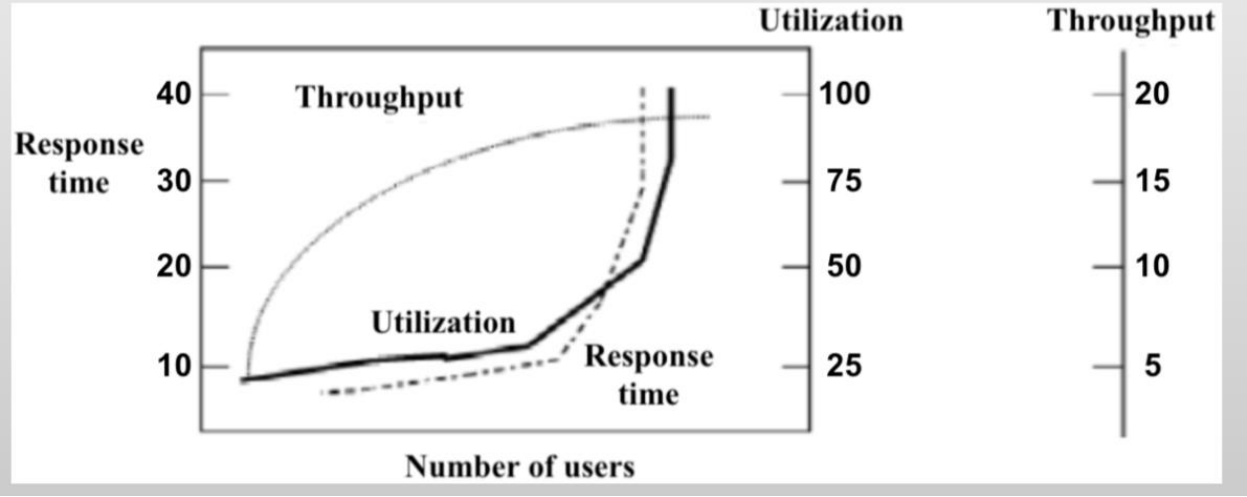
a. Show coordinate axes, scale divisions, and origin.

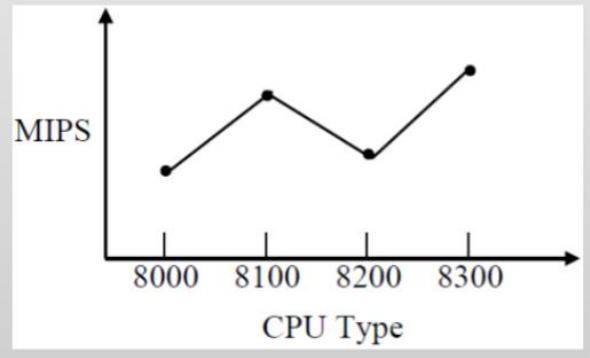
Checklist for good graphics

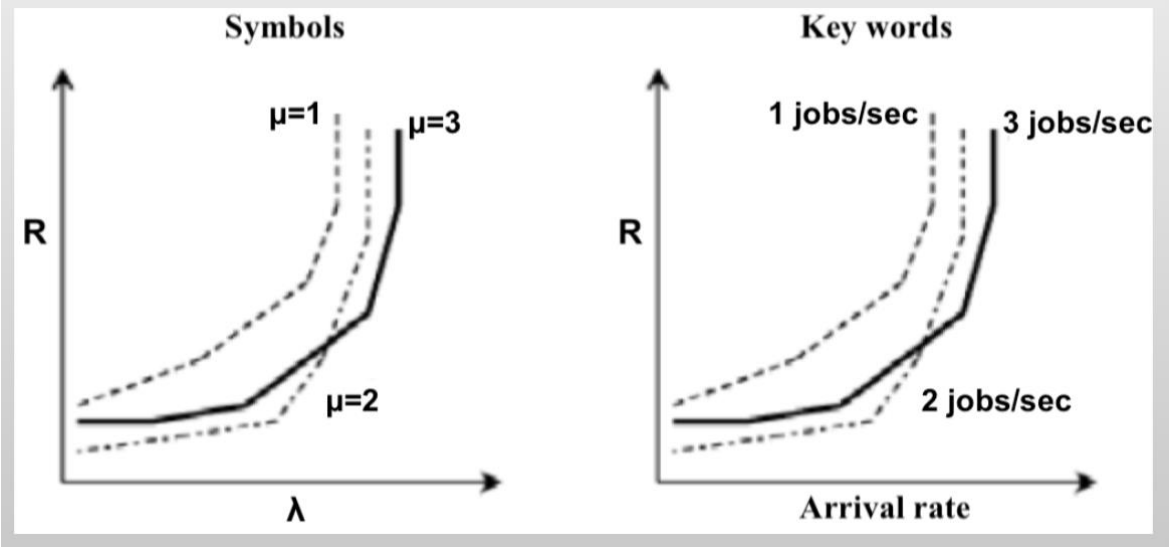
1. Are both coordinate axes shown and labeled?
2. Are the axes labels self-explanatory and concise?
3. Are the scaLes and divisions shown on both axes?
4. Are the minimum and maximum of the ranges shown on the axes appropriate to present the maximum information?
5. Is the number of curves reasonably small? (A line chart should generally have no more than six curves).
6. Do all graphs use the same scale? (Multiple scaLes on the same chart are confusing.)
7. Is there no curve that can be removed without reducing the information?
8. Are the curves on a line chart individually labeled?
9. Are the cells in a bar chart individually labeled?
10. Are all symbols on the graph accompanied by appropriate textual explanations?
11. If the curves cross, are the line patterns different to avoid confusion?
12. Are the units of measurement indicated?
13. Is the horizontal scale increasing from left to right?
14. Is the vertical scale increasing from bottom to top?
15. Are the grid lines aiding in reading the curves? (If not, the grid lines should not be shown.)
16. Does this whole chart add to information available to the reader?
17. Are the scales contiguous? (Breaks in the scale should be avoided or clearly shown.)
18. Is the order of bars in a bar chart systematic? (Alphabetic, temporal, or best-to-worst ordering is to be preferred over random placement.)
19. If the vertical axis represents a random quantity, are confidence intervals shown?
20. If the vertical axis represents a random quantity, are confidence intervals shown?
21. Are there no curves, symbols, or texts on the graph that can be removed without affecting the information? 21. Is there a title for the whole chart?
22. Is the chart title self-explanatory and concise?
23. Do the variables plotted on this chart give more information than other alternatives?
24. Does the chart clearly bring out the intended message?
25. Is the figure referenced and discussed in the text of the report?

Common mistakes

1. Too much data on a graph
    a. Rule of thumb
        i. Max: **6 lines**/curves on a line chart, **8 bars** in a bar/column chart, **8 slices** in a pie chart
        ii. Max **10 values** on an axis
2. Using symbols in place of text
3. Using line chart instead of column chart



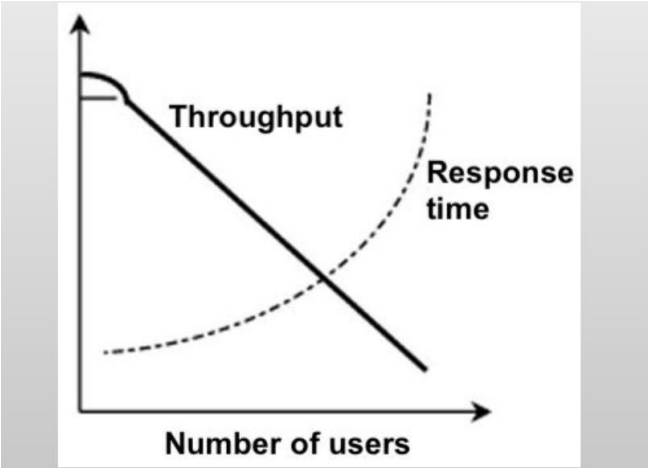


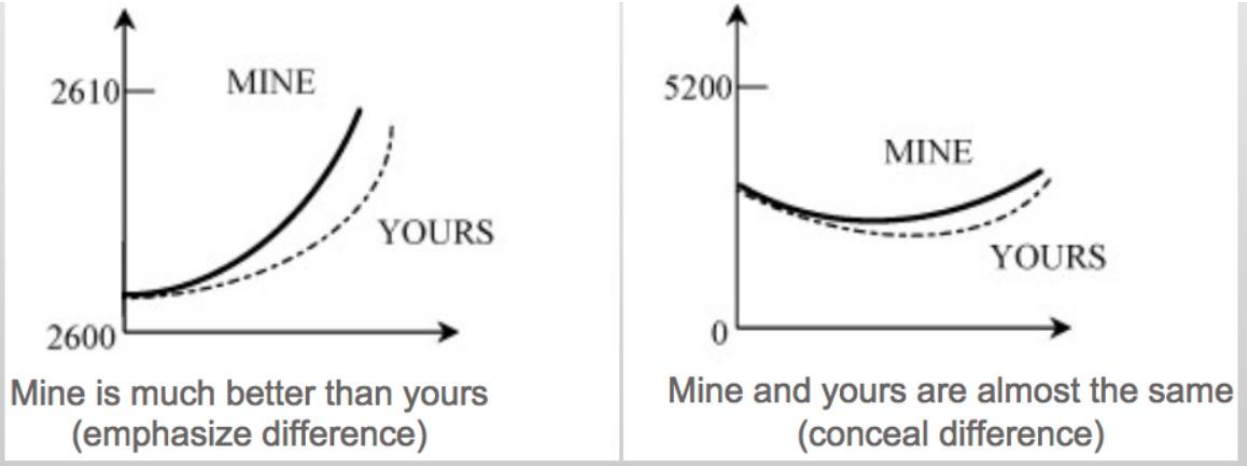


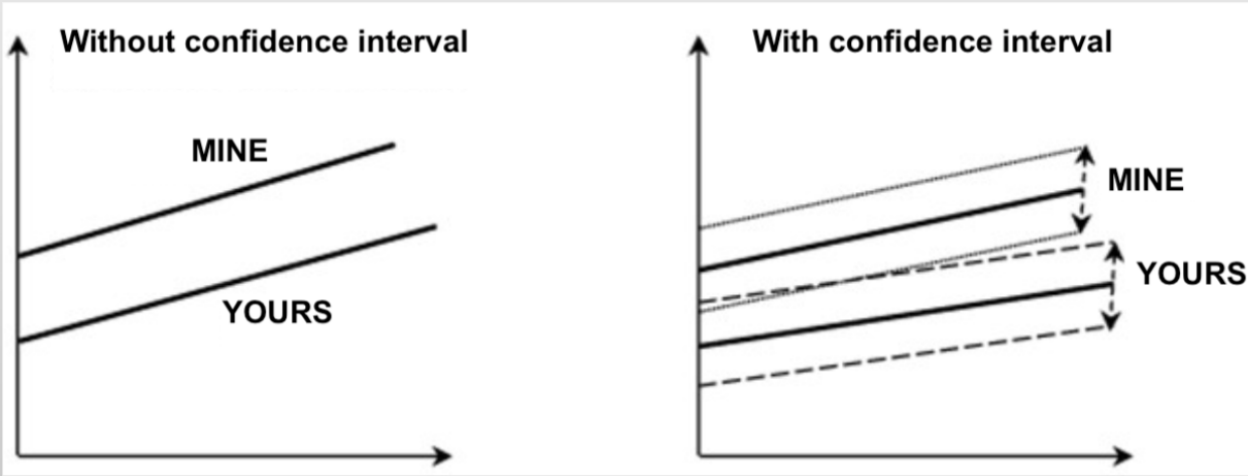
Misleading graphs

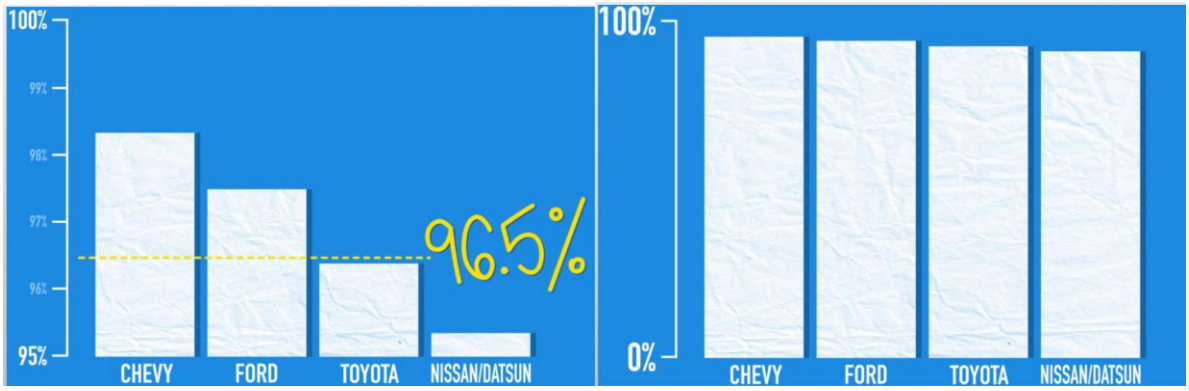
1. Using non-zero origins to emphasize the difference
2. Improper scaling
3. Plotting related metrics for 'dramatisation'

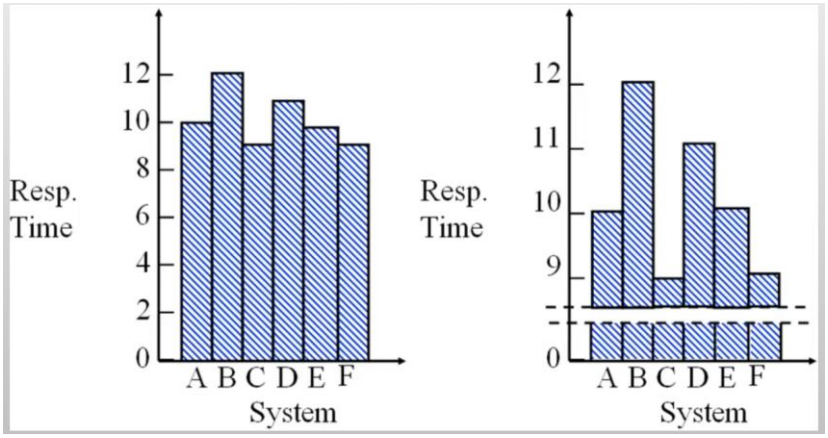
4. Omitting data
5. Improper extraction
6. Labels, numbers, scale, context



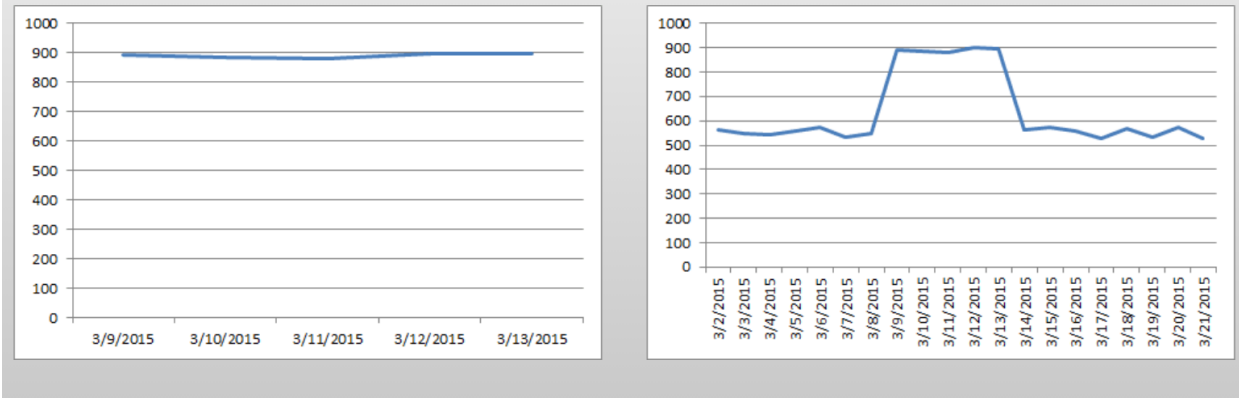


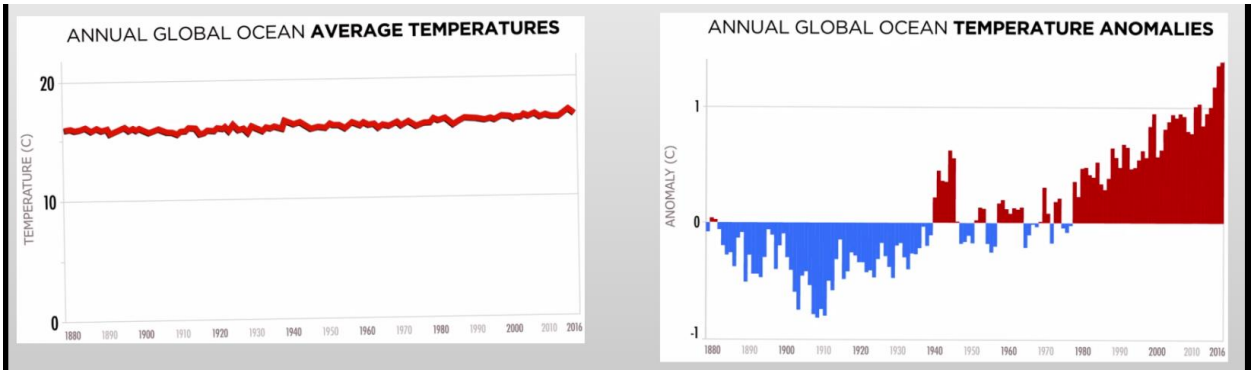


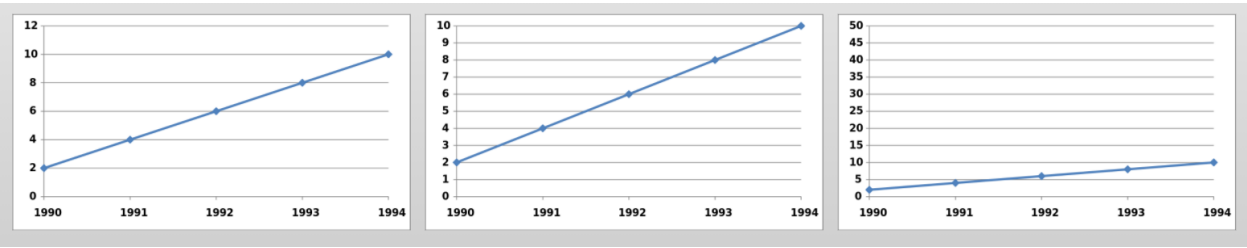

100%
99%
98%
97%
96%
95%
96.5%
CHEVY
FORD
TOYOTA
NISSAN/DATSUN
100%
0%
CHEVY
FORD
TOYOTA
NISSAN/DATSUN

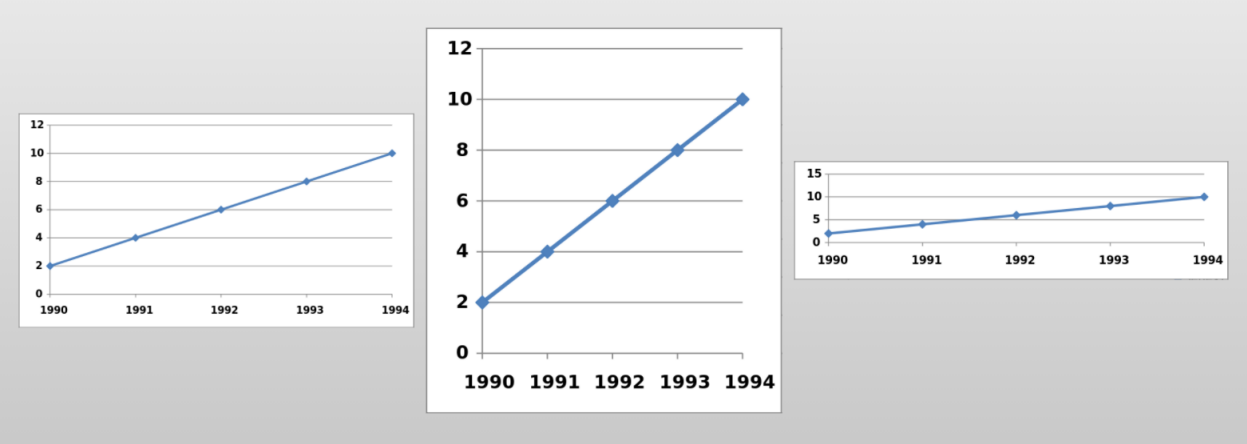

Resp. Time
12
10
8
6
4
2
0
A B C D E F
System
Resp. Time
12
11
10
9
0
A B C D E F
System

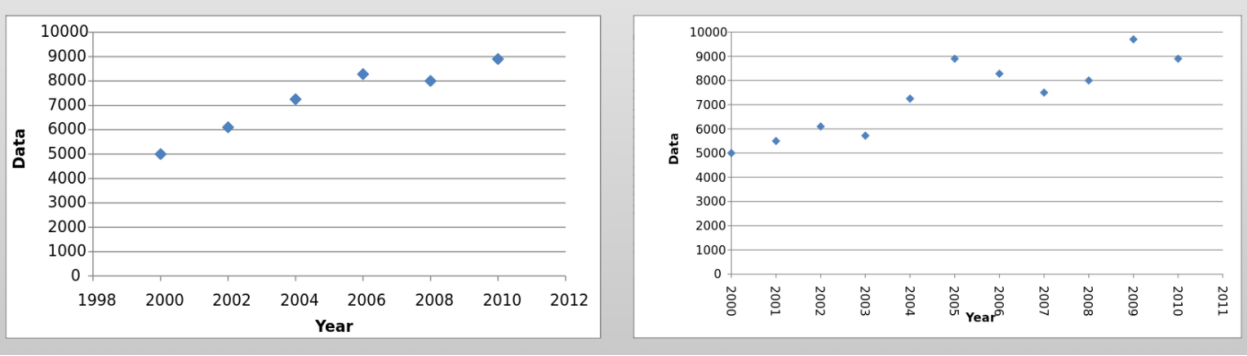

JOB LOSS BY QUARTER
15 MIL
13.5 MIL
9 MIL
7 MIL
DEC '07
SEPT '08
MAR '09
JUN '10
6
15
JOB LOSS BY QUARTER
15 MIL
7 MIL
DEC '07
SEPT '08
MAR '09
JUN '10

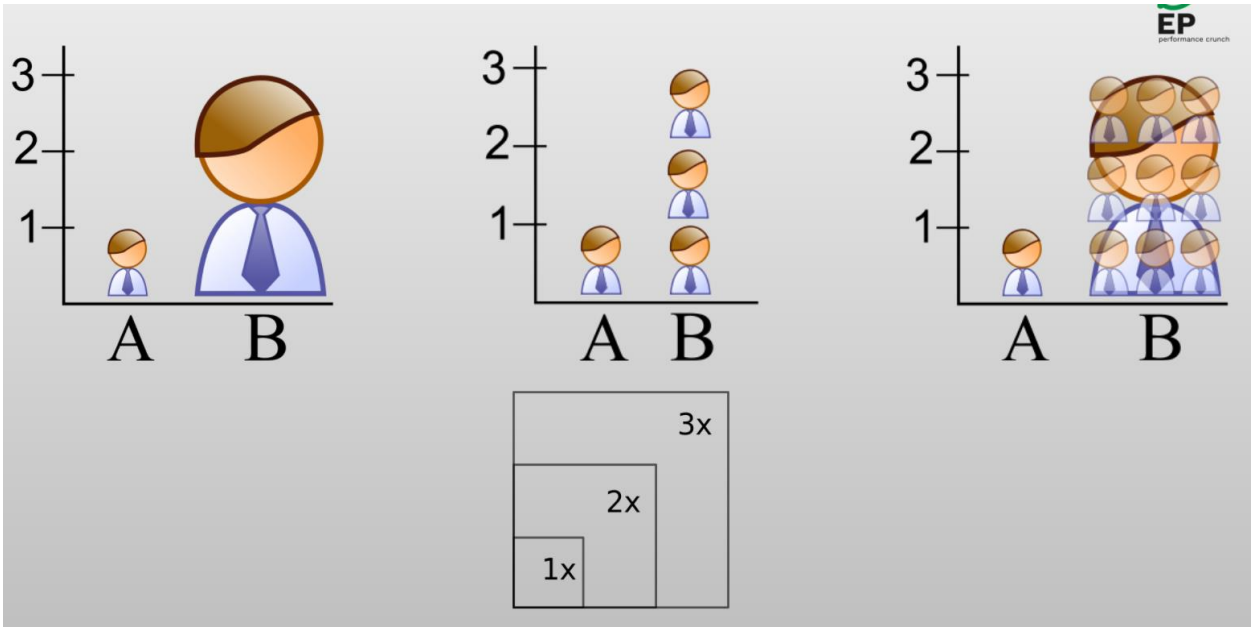

1000
900
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3/12/2015
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ANNUAL GLOBAL OCEAN AVERAGE TEMPERATURES
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ANNUAL GLOBAL OCEAN TEMPERATURE ANOMALIES
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Data
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Data
10000 9000 8000 7000 6000 5000 4000 3000 2000 1000 0
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Year



Overview

- Importance
- Guidelines / best practices
- Matplotlib – Why, Intro, More details

Plotting tools

What is a plotting tool? A plotting tool is computer software which helps to analyze and visualize data, often of a scientific nature. Using this type of software, users can generate plots of functions, data and data fits.

Software of this nature typically includes additional functionality, such as data analysis functions including curve fitting.

Plotting tools

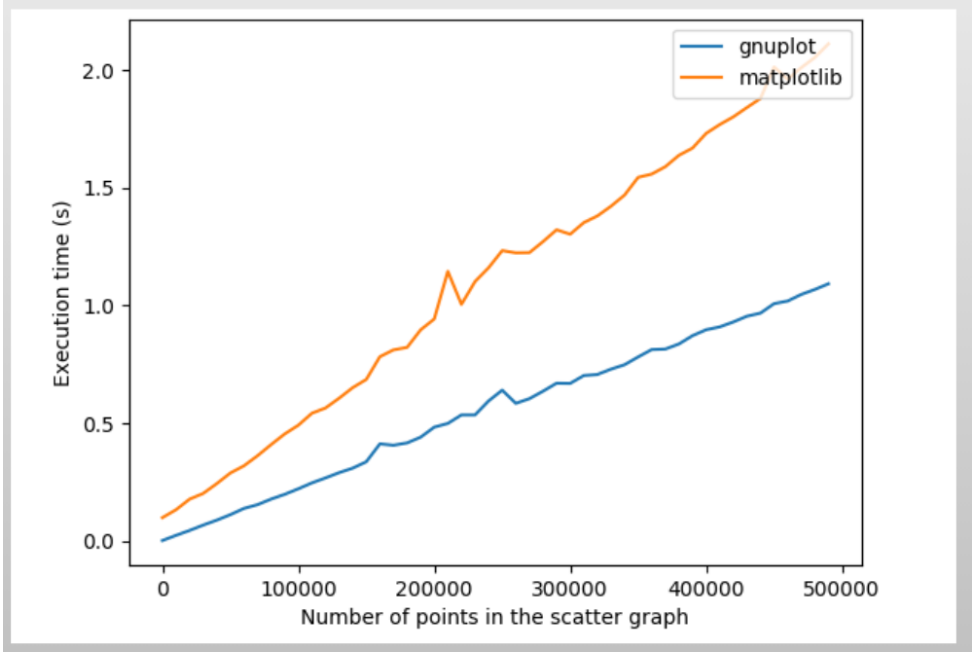
1. Online / Offline
2. Operating systems (including mobile OSs)
3. Open source / Closed source
4. Free / Paid
   Over 100 alternatives here https://en.wikipedia.org/wiki/List_of_information_graphics_software

Free, linux-based:

- **gnuptot** - Command-line driven interactive plotting program
- **Ctioga** - Command-line plotting utility

- **LabPlot** - Data plotting and function analysis tool for KDE
- **matplotlib** - Python based plotting system in a style similar to Matlab
- **Qtilnot** - Fully fledged plotting software
- **ROOT** - Solves the data analysis challenges of high-energy physics
- **SCaVis** - Produces professional graphics and has many libraries for data manipulation
- **GLE** - Graphics Layout Engine
- **PLplot** - Scientific plotting library
- **RLPLot** - Generate publication quality graphs
- **Genius** - Educational tool in mathematics and for research

Gnuplot vs. Matplotlib



Matplotlib

- **Price** - Free to download
- **Language** - Python
- **License** - Open source
- **Developer** - NumFOCUS - nonprofit organisation in the US
- **Website** - https://matplotlib.org
- **Support** - Documentation

Gnuplot

- A comprehensive library for creating static, animated, and interactive visualizations in Python
- Large user base > great documentation > actively maintained > gained popularity in recent years

Matplotlib

- MATLAB-like interface
- Low level plotting backend for virtually all plotting tools in python
    - Seaborn, pandas plotting etc.
- De-facto standard for data visualizations in data science / machine learning communities
- Interactive (ipython) or in python scripts
- 2D, 3D
- Control over all visual aspects of a plot
- Output file formats
    - PNG, EPS, SVG, JPEG, ...

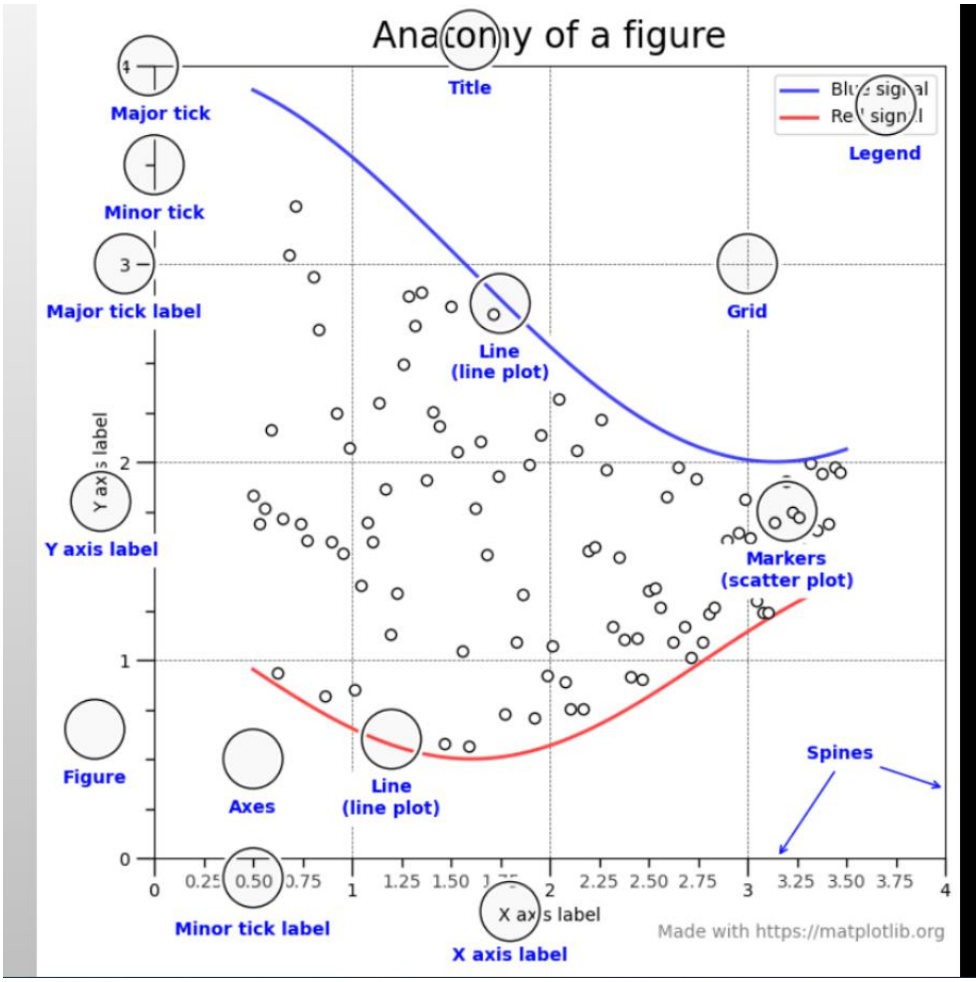
Starting up

Install matplotlib - $ pip install matplotlib

Import matplotlib - >> import matplotlib.pyplot as plt

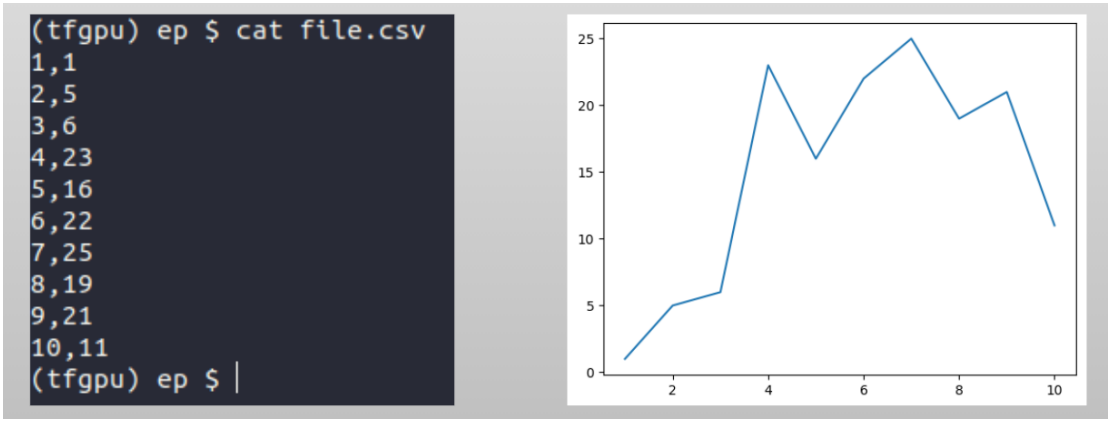
Anatomy of a figure

1. All aspects customizable
2. Axes
    a. Ranges, major / minor ticks, labels
3. Title
4. Markers
    a. Size, color, shape
5. Lines
    a. Thickness, colors, types
6. Grid
7. Legend

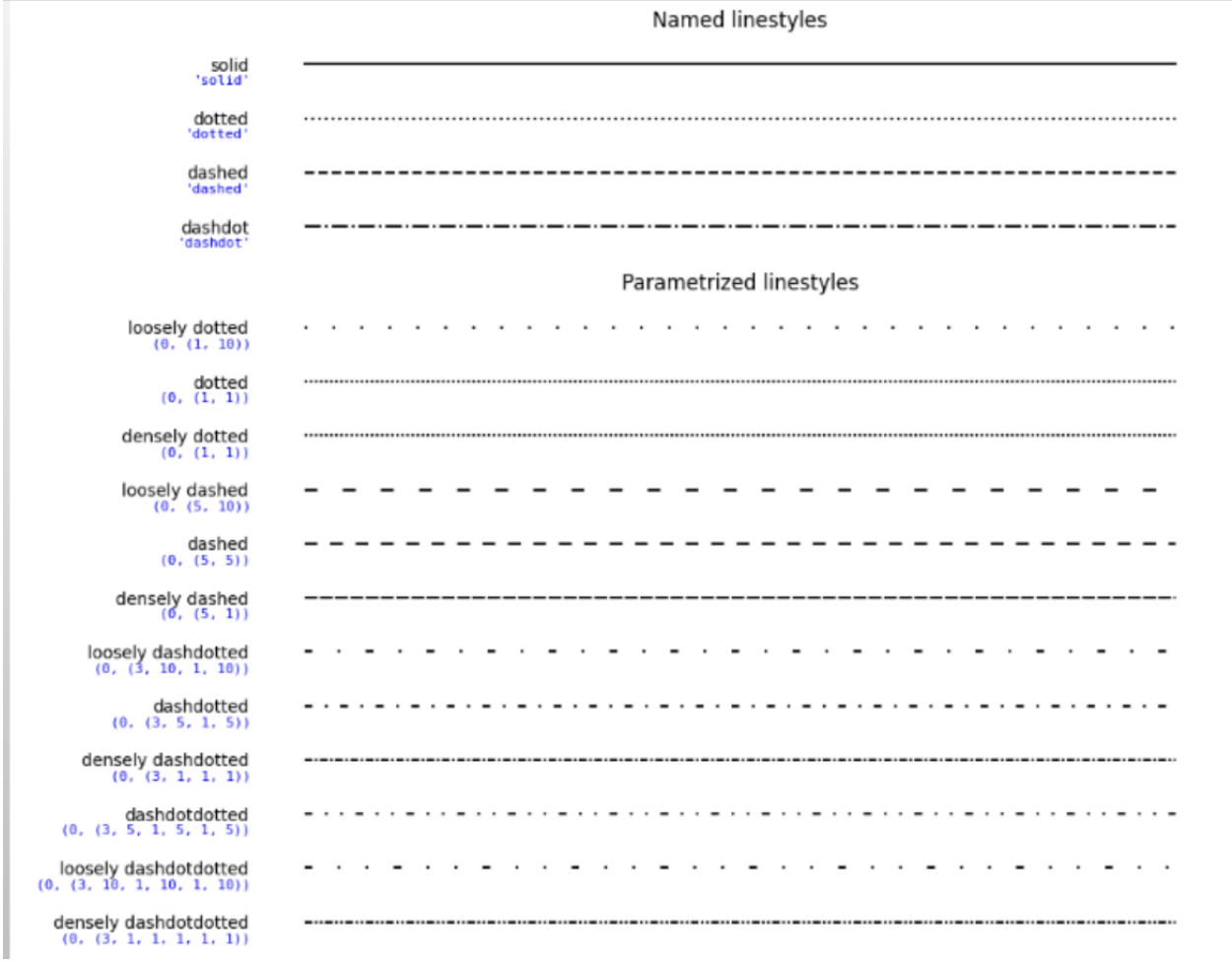


Plotting data from a file

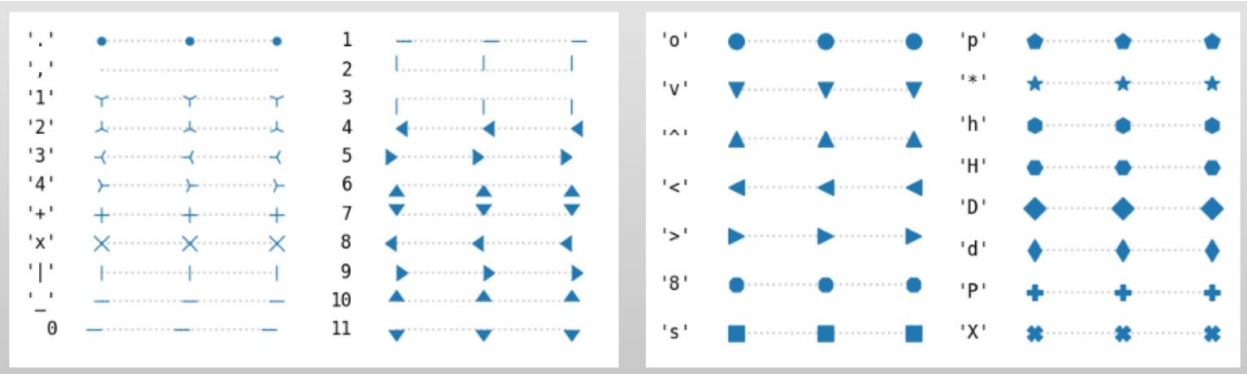
```
>> import matplotlib.•yplot as •1t
>> import numpy as np
>> data = np.genfromtxt("file.csv", delimiter=",", names=["x", "y"])
>> plt.plot(data['x'], data["])
```



Linestyles



Markers

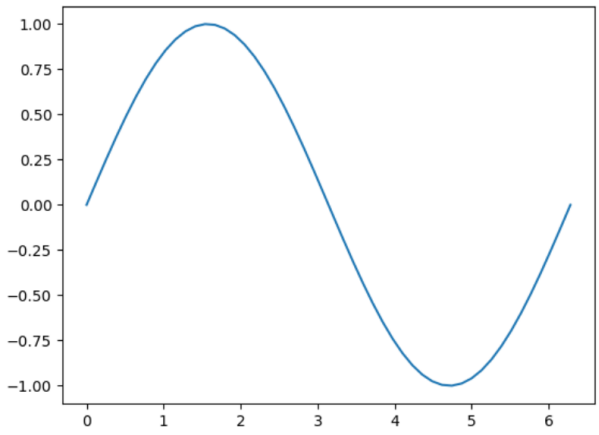


Matplotlib Showcase

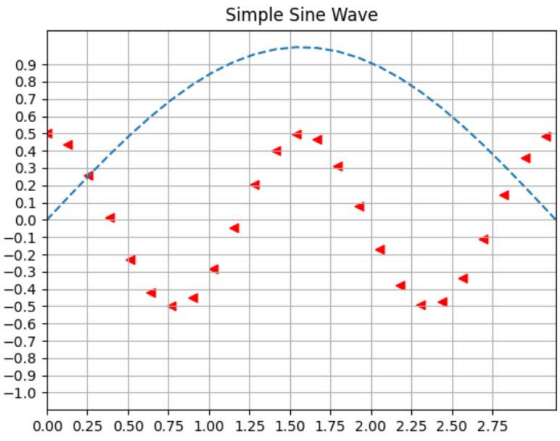
```
import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(0, 2 * np.pi)
y = np.sin(x)

plt.plot(x, y)
plt.show()
```




```
plt.plot(x, y, linestyle = '--')
plt.title('Simple Sine Wave')
plt.xlim(0, np.pi)
y2 = np.cos(2*x) ** 2 - 0.5
plt.scatter(x, y2, marker='<', c = 'r')
plt.xticks(np.arange(0, 3, 0.25))
plt.yticks(np.arange(-1, 1, 0.1))
plt.grid()
```



Subplots

```
import numpy as np
import matplotlib.pyplot as plt

x1 = np.linspace(0.0, 5.0)
x2 = np.linspace(0.0, 2.0)

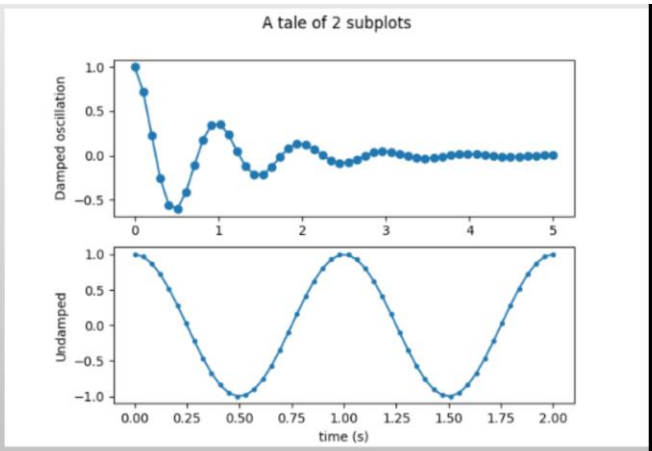
y1 = np.cos(2 * np.pi * x1) * np.exp(-x1)
y2 = np.cos(2 * np.pi * x2)

fig, (ax1, ax2) = plt.subplots(2, 1)
fig.suptitle('A tale of 2 subplots')

ax1.plot(x1, y1, 'o-')
ax1.set_ylabel('Damped oscillation')

ax2.plot(x2, y2, '.-')
ax2.set_xlabel('time (s)')
ax2.set_ylabel('Undamped')

plt.show()
```



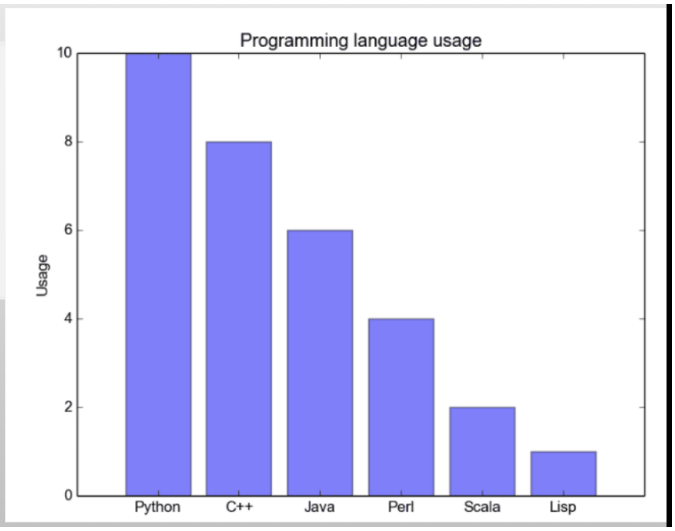
Bar chart

```
import matplotlib.pyplot as plt; plt.rcdefaults()
import numpy as np
import matplotlib.pyplot as plt

objects = ('Python', 'C++', 'Java', 'Perl', 'Scala', 'Lisp')
y_pos = np.arange(len(objects))
performance = [10,8,6,4,2,1]

plt.bar(y_pos, performance, align='center', alpha=0.5)
plt.xticks(y_pos, objects)
plt.ylabel('Usage')
plt.title('Programming language usage')

plt.show()
```



Stacked bar chart

```
import matplotlib.pyplot as plt

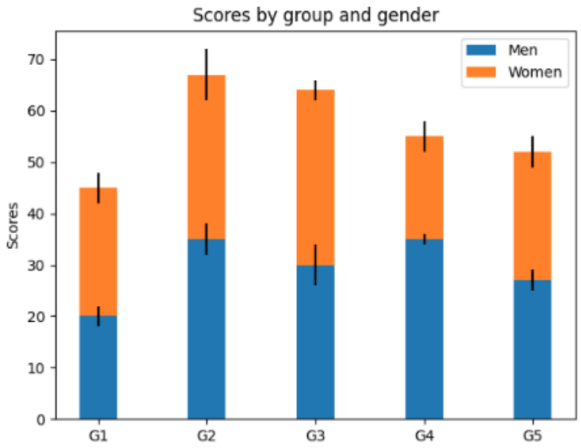

labels = ['G1', 'G2', 'G3', 'G4', 'G5']
men_means = [20, 35, 30, 35, 27]
women_means = [25, 32, 34, 20, 25]
men_std = [2, 3, 4, 1, 2]
women_std = [3, 5, 2, 3, 3]
width = 0.35       # the width of the bars: can also be len(x) sequence

fig, ax = plt.subplots()

ax.bar(labels, men_means, width, yerr=men_std, label='Men')
ax.bar(labels, women_means, width, yerr=women_std, bottom=men_means,
       label='Women')

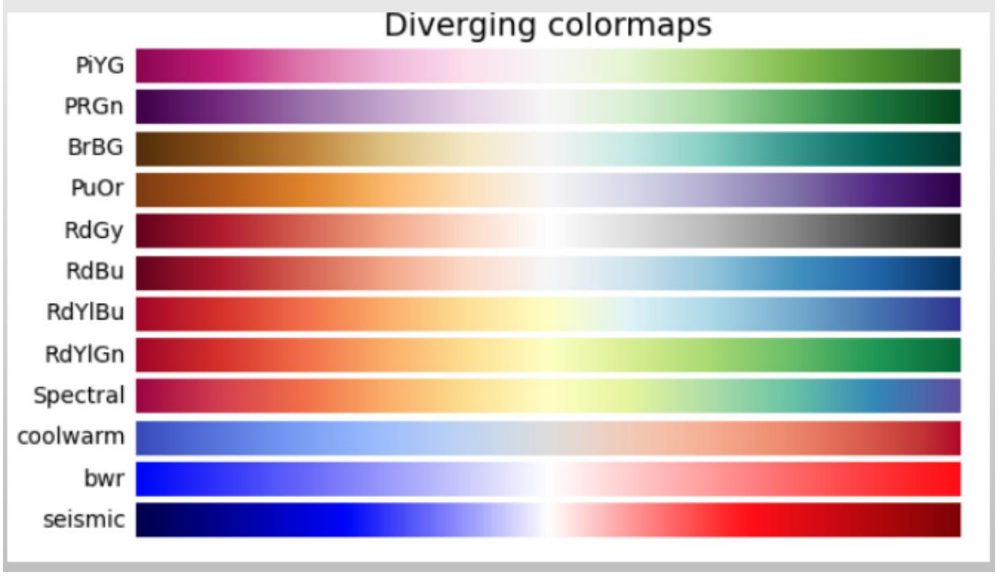
ax.set_ylabel('Scores')
ax.set_title('Scores by group and gender')
ax.legend()

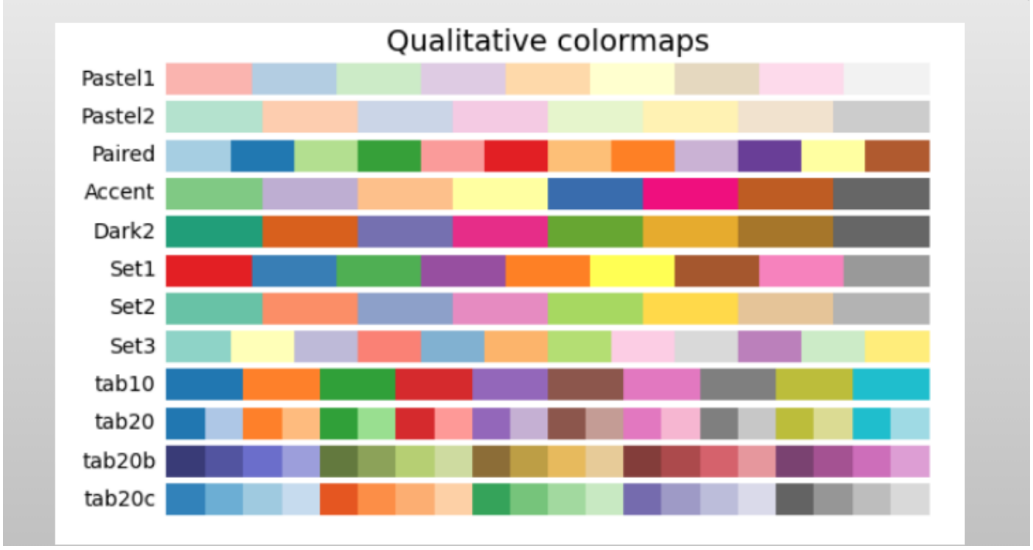
plt.show()
```



Colormaps



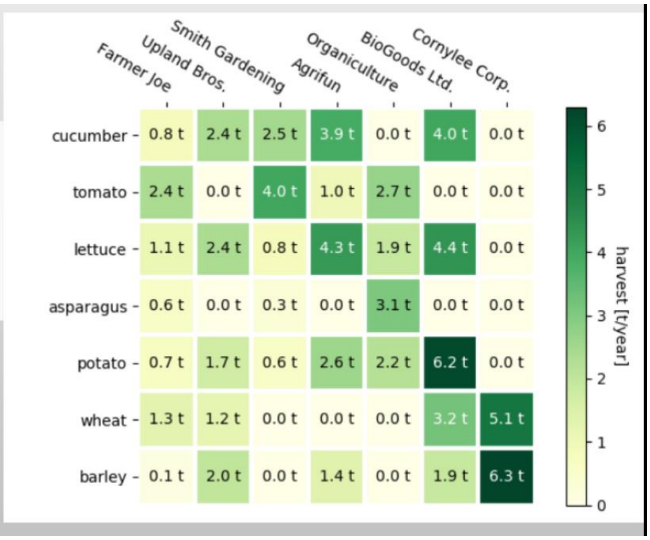




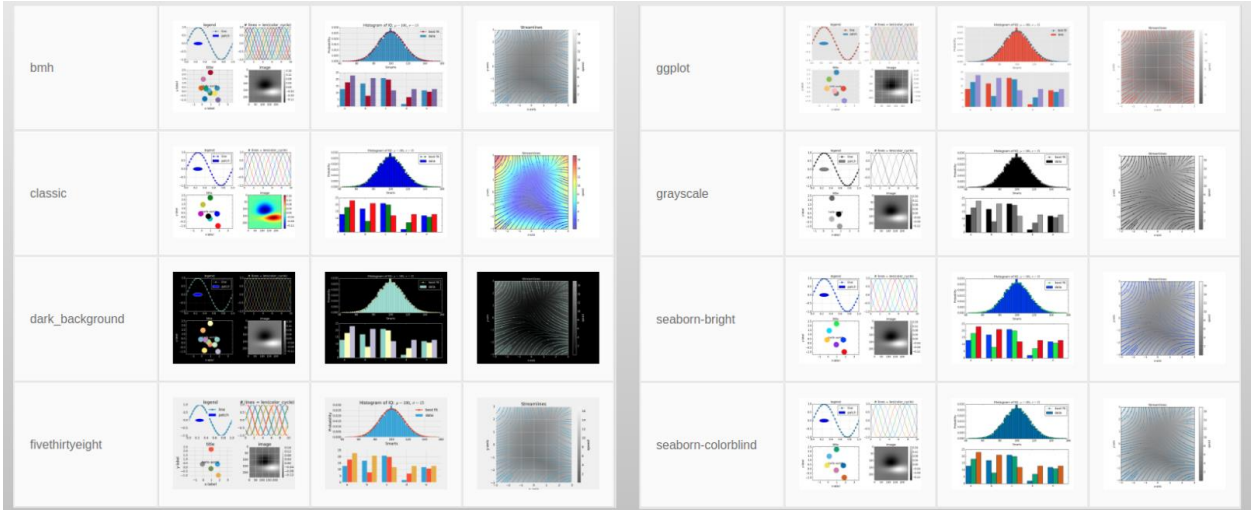

```
fig, ax = plt.subplots()

im, cbar = heatmap(harvest, vegetables, farmers, ax=ax,
                   cmap="YlGn", cbarlabel="harvest [t/year]")
texts = annotate_heatmap(im, valfmt="{x:.1f} t")

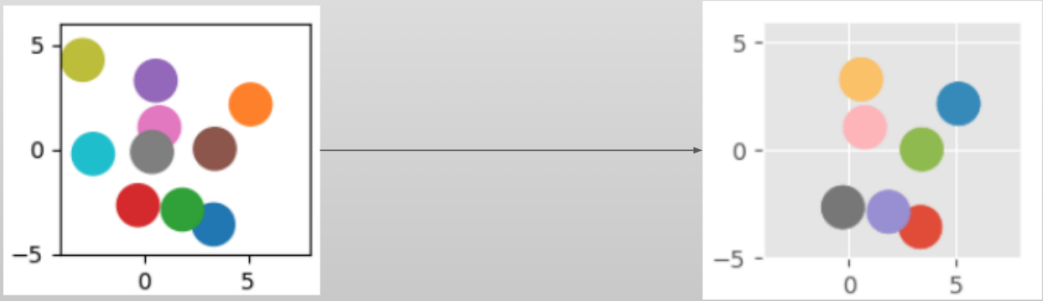
fig.tight_layout()
plt.show()
```



Styles




```
>> plt.style.use('ggplot')
```



Xkcd Style

```
with plt.xkcd():
    # Based on "Stove Ownership" from XKCD by Randall Munroe
    # https://xkcd.com/418/

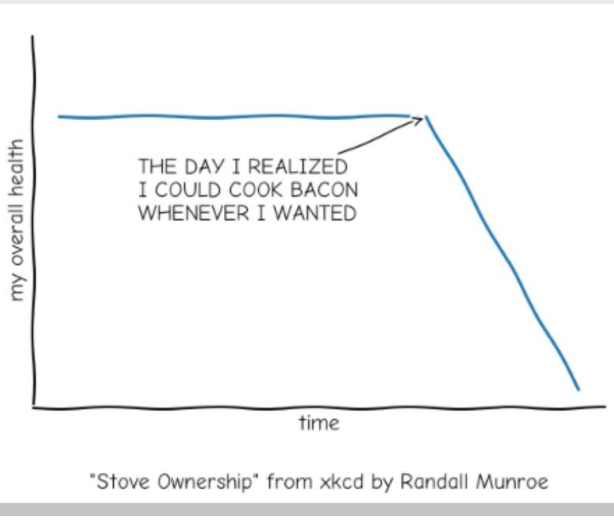
    fig = plt.figure()
    ax = fig.add_axes((0.1, 0.2, 0.8, 0.7))
    ax.spines['right'].set_color('none')
    ax.spines['top'].set_color('none')
    ax.set_xticks([])
    ax.set_yticks([])
    ax.set_ylim([-30, 10])

    data = np.ones(100)
    data[70:] -= np.arange(30)

    ax.annotate(
        'THE DAY I REALIZED\nI COULD COOK BACON\nWHENEVER I WANTED',
        xy=(70, 1), arrowprops=dict(arrowstyle='->'), xytext=(15, -10))

    ax.plot(data)

    ax.set_xlabel('time')
    ax.set_ylabel('my overall health')
    fig.text(
        0.5, 0.05,
        '"Stove Ownership" from xkcd by Randall Munroe',
        ha='center')
```



Some cool plots

Lorentz Attractor

```python
def lorenz(x, y, z, s=10, r=28, b=2.667):
    """
    Given:
       x, y, z: a point of interest in three dimensional space
       s, r, b: parameters defining the lorenz attractor
    Returns:
       x_dot, y_dot, z_dot: values of the lorenz attractor's partial
           derivatives at the point x, y, z
    """
    x_dot = s*(y - x)
    y_dot = r*x - y - x*z
    z_dot = x*y - b*z
    return x_dot, y_dot, z_dot


dt = 0.01
num_steps = 10000

# Need one more for the initial values
xs = np.empty(num_steps + 1)
ys = np.empty(num_steps + 1)
zs = np.empty(num_steps + 1)

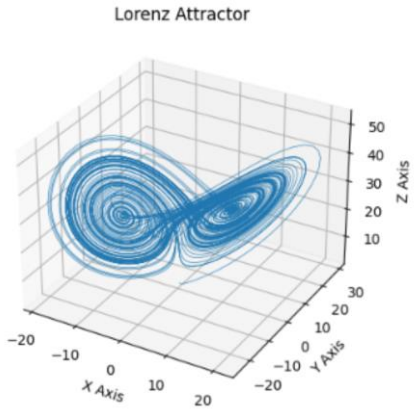
# Set initial values
xs[0], ys[0], zs[0] = (0., 1., 1.05)

# Step through "time", calculating the partial derivatives at the current point
# and using them to estimate the next point
for i in range(num_steps):
    x_dot, y_dot, z_dot = lorenz(xs[i], ys[i], zs[i])
    xs[i + 1] = xs[i] + (x_dot * dt)
    ys[i + 1] = ys[i] + (y_dot * dt)
    zs[i + 1] = zs[i] + (z_dot * dt)


# Plot
fig = plt.figure()
ax = fig.gca(projection='3d')

ax.plot(xs, ys, zs, lw=0.5)
ax.set_xlabel("X Axis")
ax.set_ylabel("Y Axis")
ax.set_zlabel("Z Axis")
ax.set_title("Lorenz Attractor")

plt.show()
```



3D Surface with polar coordinates

```python
import matplotlib.pyplot as plt
import numpy as np


fig = plt.figure()
ax = fig.add_subplot(111, projection='3d')

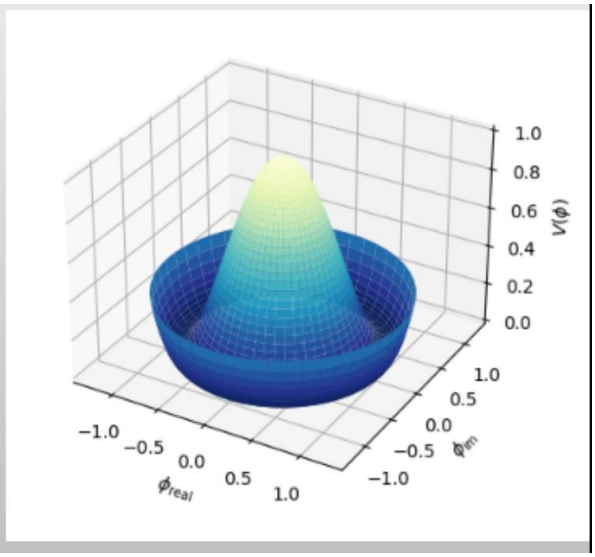
# Create the mesh in polar coordinates and compute corresponding Z.
r = np.linspace(0, 1.25, 50)
p = np.linspace(0, 2*np.pi, 50)
R, P = np.meshgrid(r, p)
Z = ((R**2 - 1)**2)

# Express the mesh in the cartesian system.
X, Y = R*np.cos(P), R*np.sin(P)

# Plot the surface.
ax.plot_surface(X, Y, Z, cmap=plt.cm.YlGnBu_r)

# Tweak the limits and add latex math labels.
ax.set_zlim(0, 1)
ax.set_xlabel(r'$\phi_\mathrm{real}$')
ax.set_ylabel(r'$\phi_\mathrm{im}$')
ax.set_zlabel(r'$V(\phi)$')

plt.show()
```



Costum hillshading in a 3D surface plot

```
from matplotlib import cbook
from matplotlib import cm
from matplotlib.colors import LightSource
import matplotlib.pyplot as plt
import numpy as np

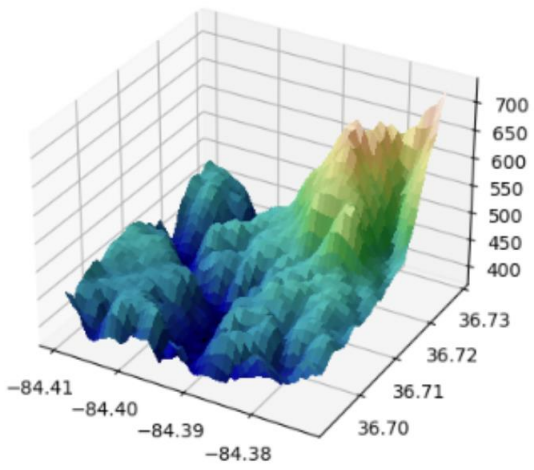
# Load and format data
dem = cbook.get_sample_data('jacksboro_fault_dem.npz', np_load=True)
z = dem['elevation']
nrows, ncols = z.shape
x = np.linspace(dem['xmin'], dem['xmax'], ncols)
y = np.linspace(dem['ymin'], dem['ymax'], nrows)
x, y = np.meshgrid(x, y)

region = np.s_[5:50, 5:50]
x, y, z = x[region], y[region], z[region]

# Set up plot
fig, ax = plt.subplots(subplot_kw=dict(projection='3d'))

ls = LightSource(270, 45)
# To use a custom hillshading mode, override the built-in shading and pass
# in the rgb colors of the shaded surface calculated from "shade".
rgb = ls.shade(z, cmap=cm.gist_earth, vert_exag=0.1, blend_mode='soft')
surf = ax.plot_surface(x, y, z, rstride=1, cstride=1, facecolors=rgb,
                       linewidth=0, antialiased=False, shade=False)

plt.show()
```



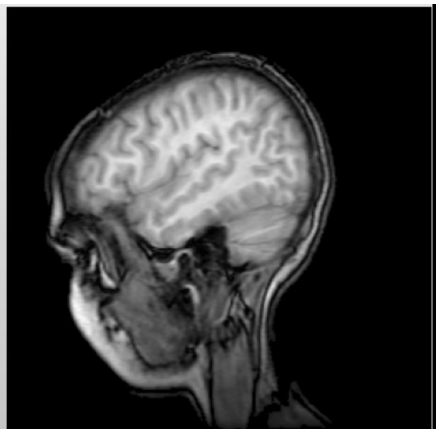
MRI Visualization

```
import matplotlib.pyplot as plt
import matplotlib.cbook as cbook
import numpy as np

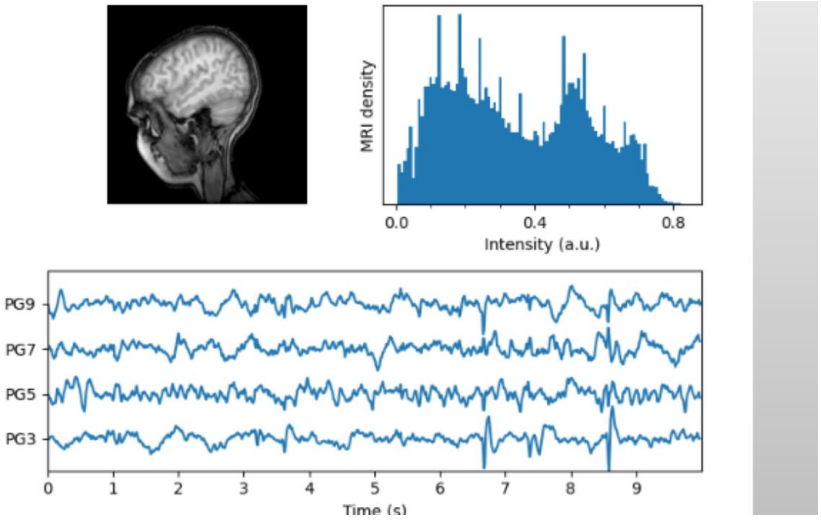

# Data are 256x256 16 bit integers.
with cbook.get_sample_data('s1045.ima.gz') as dfile:
    im = np.frombuffer(dfile.read(), np.uint16).reshape((256, 256))

fig, ax = plt.subplots(num="MRI_demo")
ax.imshow(im, cmap="gray")
ax.axis('off')

plt.show()
```



MRI Visualization with EEG



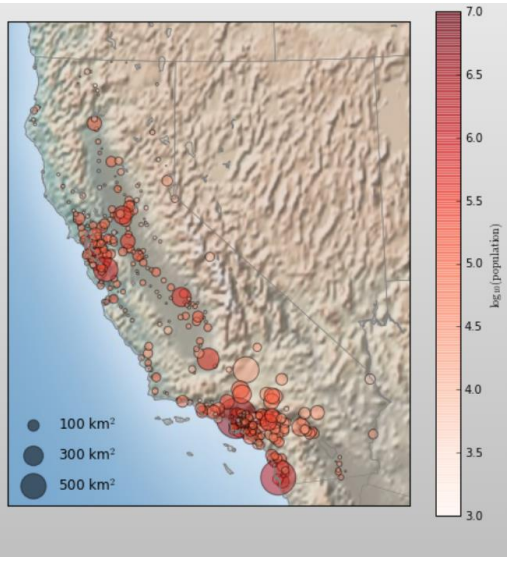
Plotting on a map with Basemap

```
# 1. Draw the map background
fig = plt.figure(figsize=(8, 8))
m = Basemap(projection='lcc', resolution='h',
            lat_0=37.5, lon_0=-119,
            width=1E6, height=1.2E6)
m.shadedrelief()
m.drawcoastlines(color='gray')
m.drawcountries(color='gray')
m.drawstates(color='gray')

# 2. scatter city data, with color reflecting population
# and size reflecting area
m.scatter(lon, lat, latlon=True,
          c=np.log10(population), s=area,
          cmap='Reds', alpha=0.5)

# 3. create colorbar and legend
plt.colorbar(label=r'$\log_{10}({\rm population})$')
plt.clim(3, 7)

# make legend with dummy points
for a in [100, 300, 500]:
    plt.scatter([], [], c='k', alpha=0.5, s=a,
                label=str(a) + ' km$^2$')
plt.legend(scatterpoints=1, frameon=False,
           labelspacing=1, loc='lower left');
```



Conclusion

- Always consider the guidelines for good graphs (most importantly -easy to read / understand, self-contained).
- The lecture presented some of the basic utility of matplotlib, but this is just scratching the surface.
- Any 2D or 3D plots you have seen can be rendered in very high quality using matplotlib.

Curs 5 – Summarizing Measured Data

Overview

- Ratio games
- Summarizing measured data

Ratio games

Example

- Mice trials of 2 types of cancer medicine: **M1** and **M2**.
    - Stats for the last 1000 mices.

	M1	M2
1000	**900**	800
Advanced Stage	100/30	400/210
	30%	**52.5%**
Early Stage	900/870	600/590
	96.6%	**98.3%**

Example

Bench-mark	System 6502	System 8080
Block	41.16	51.50
Sieve	63.17	48.08
Sum	104.33	99.58
Avg	52.17	49.79

System 6502	System 8080
1.00	1.25
1.00	0.76
2.00	2.01
1.00	1.01

System 6502	System 8080
0.80	1.00
1.31	1.00
2.11	2.00
1.06	1.00

- Ratio of totals
    - 6502 is worse - takes 4.7% more time
- 6502 as base
    - 6502 is better - takes 1% less time
- 8080 as base
    - 6502 is worse - takes 6% more time

System	Throughput	Response	Power
A	10	2	5
B	4	1	4

- Throughput: A is better
- Response: A is worse
- Power: A is better

Alternative	Without	With	Ratio
A on X	2	4	2.00
B on Y	3	5	1.66

- Incomparable bases: need to try both on the same machine

System A:	Test	Total	Pass	% Pass
	1	300	60	20%
	2	50	2	4%
	Total	350	62	20.6%
System B:	Test	Total	Pass	% Pass
	1	32	8	25%
	2	500	40	8%
	Total	532	48	9%

- System B is better on both tests
- System A is better overall

Guidelines

1. System better on all benchmarks
    - contradicting conclusions cannot be drawn by any ratio game technique
2. System better on all benchmarks
    - a better relative performance can be shown by selecting appropriate base
3. System better on some benchmarks
    - contradicting conclusions can be drawn in some cases (not in all cases)
4. Performance metric is an LB metric
    - better to use your system as the base
5. Performance metric is an HB metric
    - better to use your opponent as the base

6. Benchmarks that perform better on your system should be elongated and those that perform worse should be shortened

Takeaway

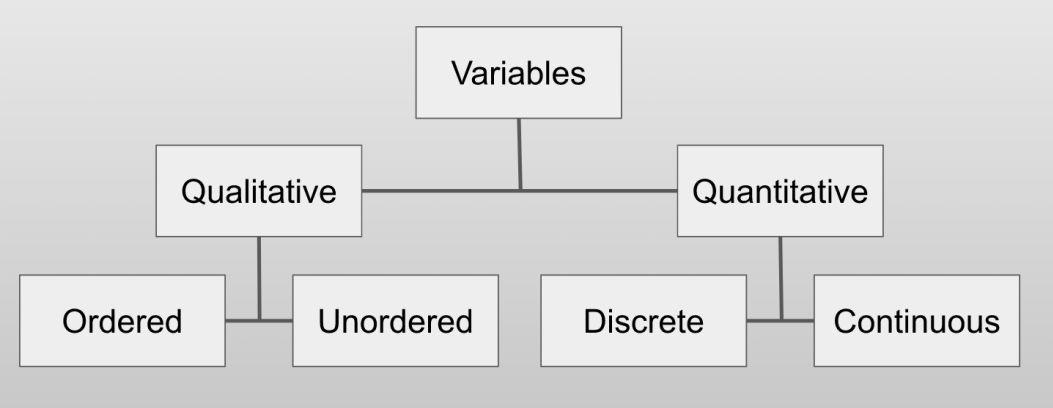
- Ratio games arise from the use of incomparable bases

Summarising measured data

Overview

- Basic probability and statistics concepts: CDF, PDF, PMF, Mean, Variance, CoV, Normal distribution
- Summarising data by a single number: mean, median, mode, arithmetic/geometric/harmonic means
- Summarising variability: range, variance, percentiles, quartiles

Recap: Types of variables



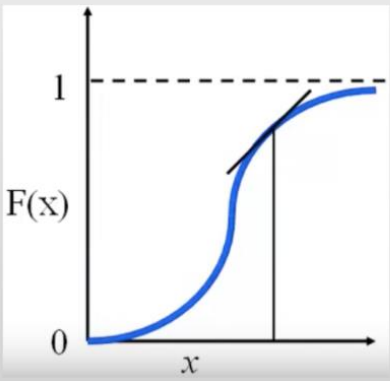
Basic concepts

- Independent events
  - The occurence of one event does not affect the probability of the other event.
  - Examples
    - Successive throws of a coin - independent
    - Price of a stock on successive days – dependent
- Random variable
  - Takes one of a specified set of values with a specified probability.
  - Examples
    - Temperature
    - Execution time

CDF, PDF, PMF

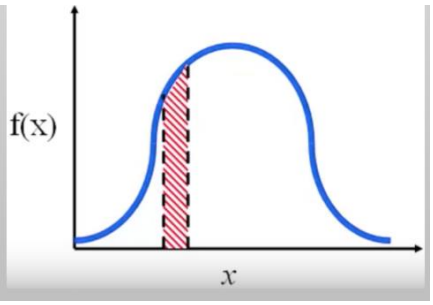
- Cumulative Distribution Function
    - Starts at 0, ends at 1

$$F(x) = P(X \leq x)$$
$$F_x(a) = P(x \leq a)$$



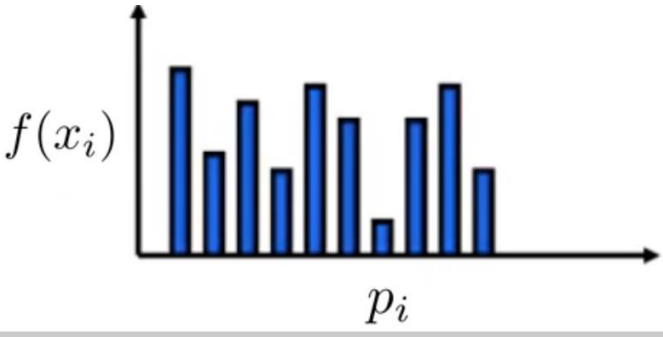
- Probability Density Function
    - Starts at 0, ends at 0.

$$f(x) = \frac{dF(x)}{dx}$$
$$P(a < x < b) = F(b) - F(a) = \int_a^b f(x)dx$$



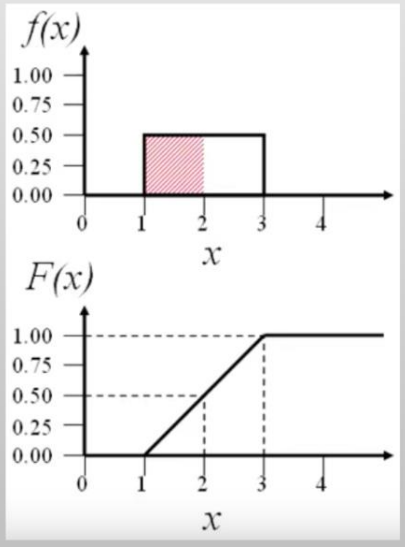
- Probability Mass Function
    - For discrete random variables

$$f(x) = P(X = x)$$
$$f(x_i) = p_i$$



Example

- Response time – uniformly distributed between 1 and 3 seconds



$$f(x) = \frac{1}{2}, \quad 1 \leq x \leq 3$$

$$P(x \leq 2) = 0.5$$

$$F(x) = \int_{-\infty}^{x} f(x)dx$$

Mean, Variance, CoV

- Mean

$$\mu = E(x) = \sum_{i=1}^{n} p_i x_i = \int_{-\infty}^{+\infty} x f(x)dx$$

- Variance
  - Expected value of the square of the distance between x and its mean.

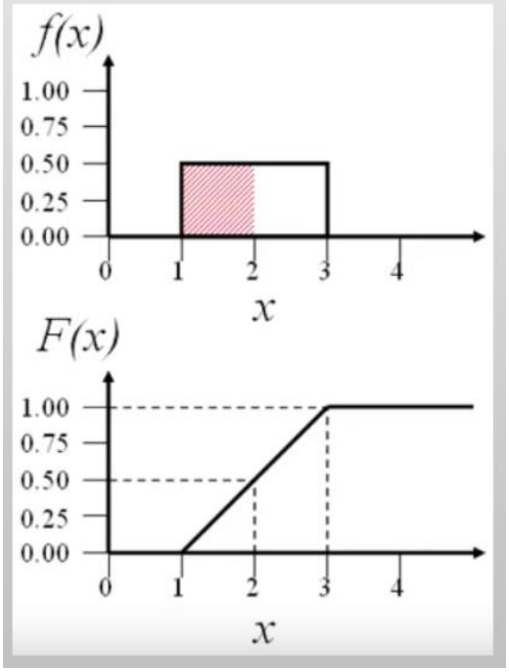
$$Var(x) = \sigma^2 = E[(x-\mu)^2] = \sum_{i=1}^{n} p_i (x_i - \mu)^2 = \int_{-\infty}^{+\infty} (x_i - \mu)^2 f(x)dx$$

- Coefficient of Variation

$$C.O.V. = \frac{Standard Deviation}{Mean} = \frac{\sigma}{\mu}$$

Example

$$f(x) = \frac{1}{2}, \quad 1 \le x \le 3$$



- **Mean**

$$\mu = E(x) = \int_{-\infty}^{+\infty} x f(x) dx = \int_{1}^{3} x \frac{1}{2} dx = \frac{x^2}{4} = \frac{9}{4} - \frac{1}{4} = 2$$

- **Variance**

$$Var(x) = \int_{-\infty}^{+\infty} (x-\mu)^2 f(x) dx = \int_{1}^{3} (x-2)^2 \frac{1}{2} dx = \frac{(x-2)^3}{6} = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

- **Standard deviation**

$$\sigma = \frac{1}{\sqrt{3}}$$

- **Coefficient of Variation**

$$C.O.V. = \frac{StandardDeviation}{Mean} = \frac{\sigma}{\mu} = \frac{1}{2\sqrt{3}}$$

Covariance and correlation

Covariance

$$Cov(x) = \sigma_{xy} = E[(x-\mu_x)(y-\mu_y)] = E(xy) - E(x)E(y)$$

- Independent variables => covariance is 0

$$E(xy) = E(x)E(y)$$

- Reverse not true

Correlation

- Normalized value of covariance
- Always between =1 and 1
- Dependence implies correlation
- Reverse NOT true

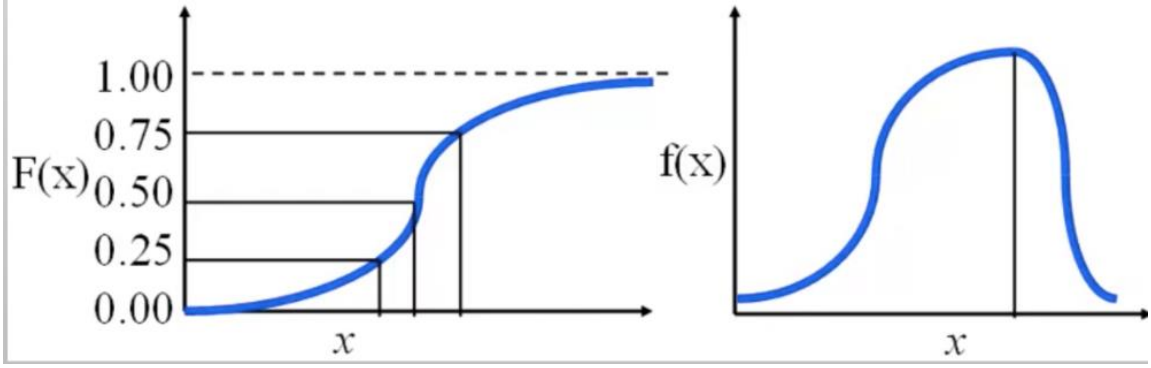
$$Correlation(x, y) = \rho_{xy} = \frac{\sigma_{xy}^2}{\sigma_x \sigma_y}$$

Quantiles, Median, Mode

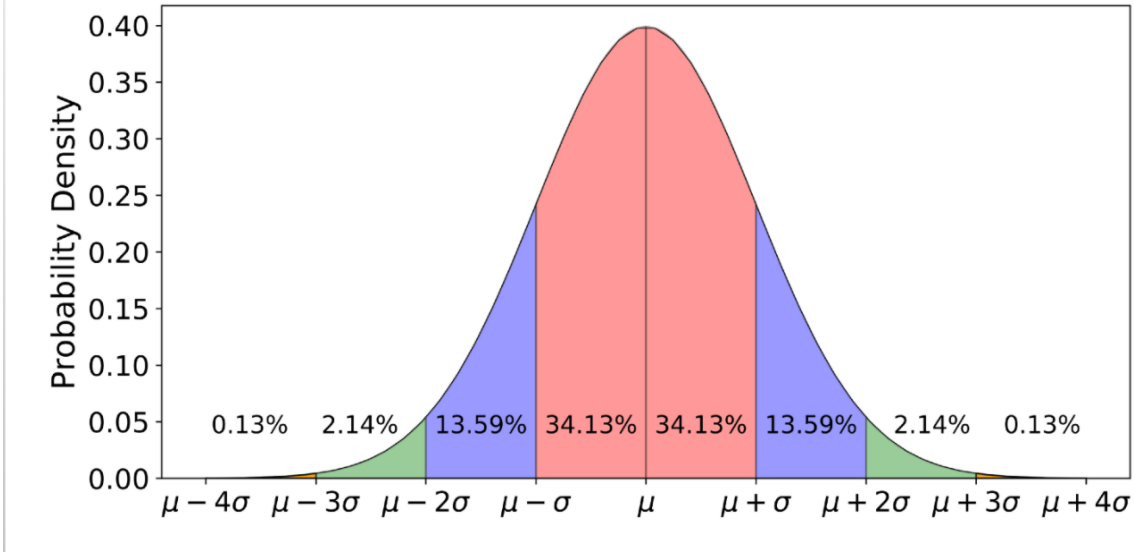
- Quantile
  - The x value at which the CDF takes a value a is called the a-quantile or 100a-percentile.

$$P(x \leq x_\alpha) = F(x_\alpha) = \alpha$$

- Median
  - 50-percentile or 0.5-quantile of a random variable
- Mode
  - The most likely value (x that has the highest probability; x at which PDF is maximum)



68-95-99.7 rule



Normal distribution

- The sum of a large number of independent observations from any distribution has a normal distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty \leq x \leq +\infty$$

- Normal variate $N(\mu, \sigma)$
- Standard normal distribution $N(0, 1)$

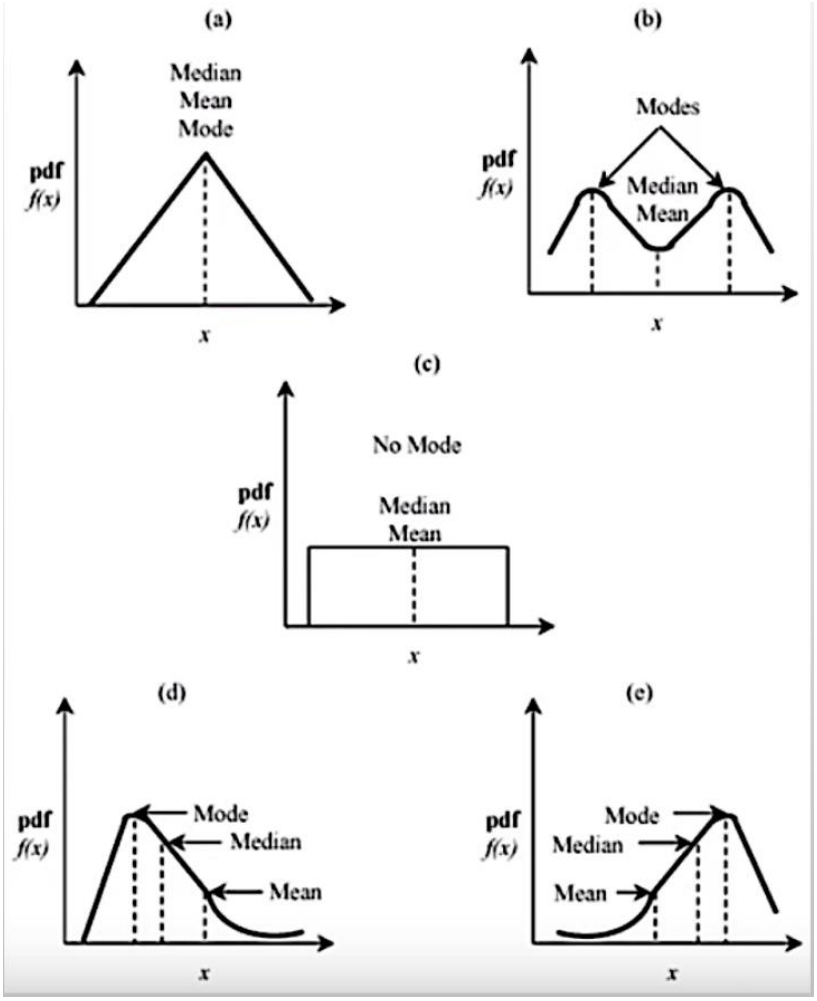
Summarising data by a single number

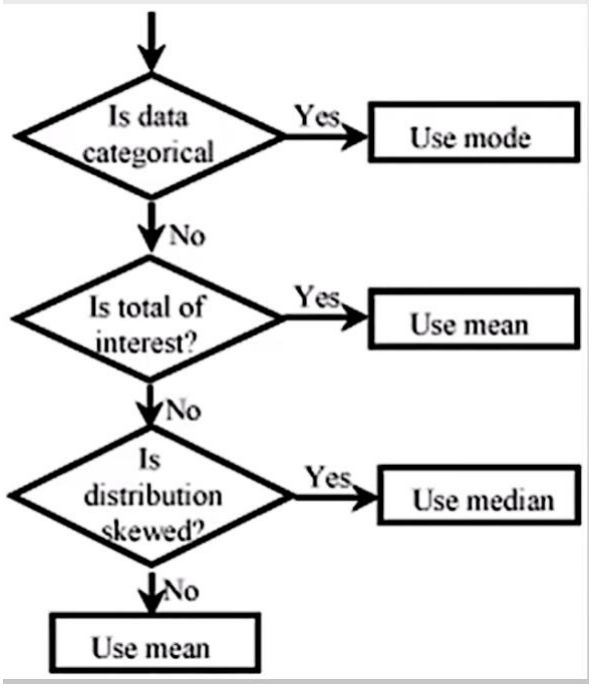
Indices of central tendencies

- Mean
- Median
- Mode

Population vs Sample

- Sample Mean
  - Sum of all observations divided by number of observations
- Sample Median
  - Sort obs in increasing order. Middle of the series. Even number mean of middle 2 values.
- Sample Mode
  - Histogram midpoint where the histogram peaks (most frequent category)


(a)
Median
Mean
Mode
pdf
f(x)
x
(b)
Modes
Median
Mean
pdf
f(x)
x
(c)
No Mode
Median
Mean
pdf
f(x)
x
(d)
Mode
Median
Mean
pdf
f(x)
x
(e)
Mode
Median
Mean
pdf
f(x)
x


Is data categorical
Yes
Use mode
No
Is total of interest?
Yes
Use mean
No
Is distribution skewed?
Yes
Use median
No
Use mean

Examples

- Most used resource in a system
- Interarrival time
- Load on a computer
- Average configuration
    - Number of devices
    - Memory size
    - Number of processors

Misuses of means

1. Using mean of significantly different values
   (10 + 1000)/2 = 505
2. Using mean without regard to the skewness of distribution

System A	System B
10	5
9	5
11	5
10	4
10	31
Sum=50	Sum=50
Mean=10	Mean=10
Typical=10	Typical=5

3. Multiplying means to get the mean of a product

   $$E(xy) \neq E(x)E(y)$$

   Ex. Time sharing system
    - Avg no of users – 23
    - Avg no of processes per user – 2
    - What is the avg no of processes?
        - is it 46?
        - No!
        - The no of processes a user spawns depends on the load on the system
4. Taking a mean of a ratio with different bases

Geometric mean

$$\dot{x} = \left(\prod_{i=1}^{n} x_i\right)^{\frac{1}{n}}$$

- Used if product of observations is of interest

Example

- Performance improvements in 7 layers

Protocol Layer	Performance Improvement
7	18%
6	13%
5	11%
4	8%
3	10%
2	28%
1	5%

- Average improvement per layer

$$\left\{(1.18)(1.13)(1.11)(1.08)(1.10)(1.28)(1.05)\right\}^{\frac{1}{7}} - 1 = 0.13$$

Examples of multiplicative metrics

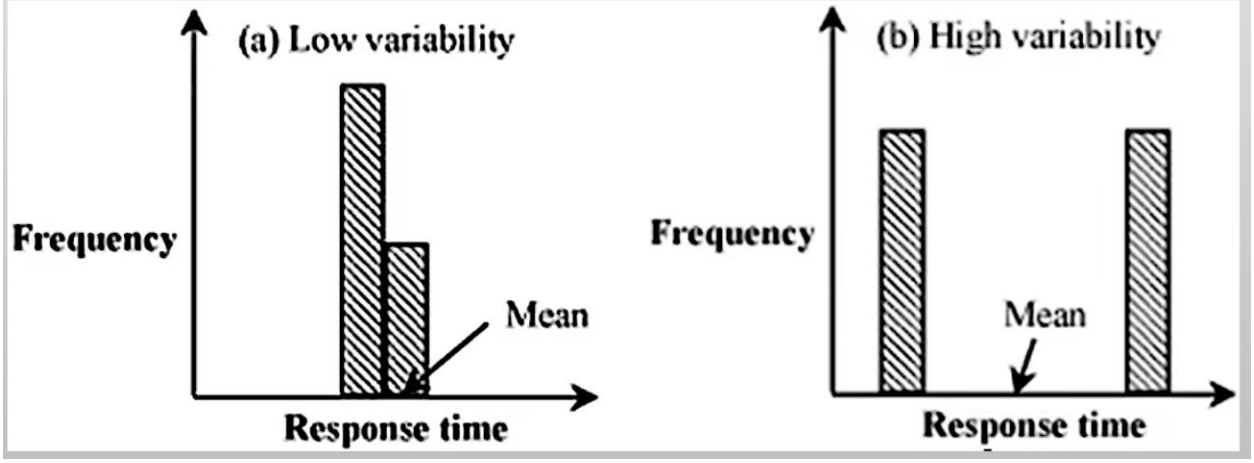
- Cache hit ratios over several levels of cache
- Cache miss ratios
- Percentage performance improvement between successive versions
- Avg error rate per hop (on a multi-hop path in a network)

Harmonic mean

$$\ddot{x} = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \ldots + \frac{1}{x_n}}$$

- Used when an arithmetic mean can be justified for $1/x_i$

Summarising variability



Indices of dispersion

- Range
  - Minimum and maximum
- Variance or standard deviation
- 10- and 90- percentiles
- Semi inter-quantile range
- Mean absolute deviation

Range

- Range = Max – Min
- Larger => higher variability

- In most cases, not very useful
    - Min often comes out to zero
    - Max often comes out to an outlier
- Unless the variable is bounded, the Max increases with the no of observations, Min decreases with the no of obs. No stable point that gives good indication of actual range.
- **Useful if there is a reason to believe that the variable is bounded**

Variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \quad where \quad \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

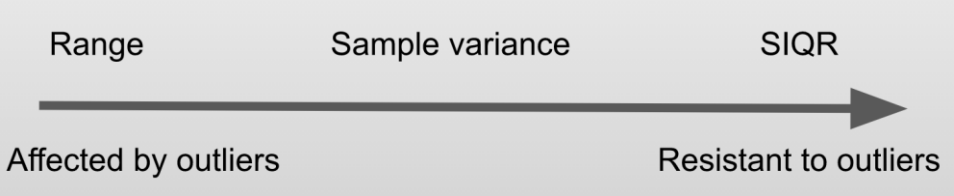
- The divisor for s2 is n-1 and not n
- The number of independent terms in a sum is also called degrees of freedom
- Expressed in units (square of units of observations)
- What is preferred to variance?
    - Variance < Std Dev < Coefficient of Variance

Percentiles

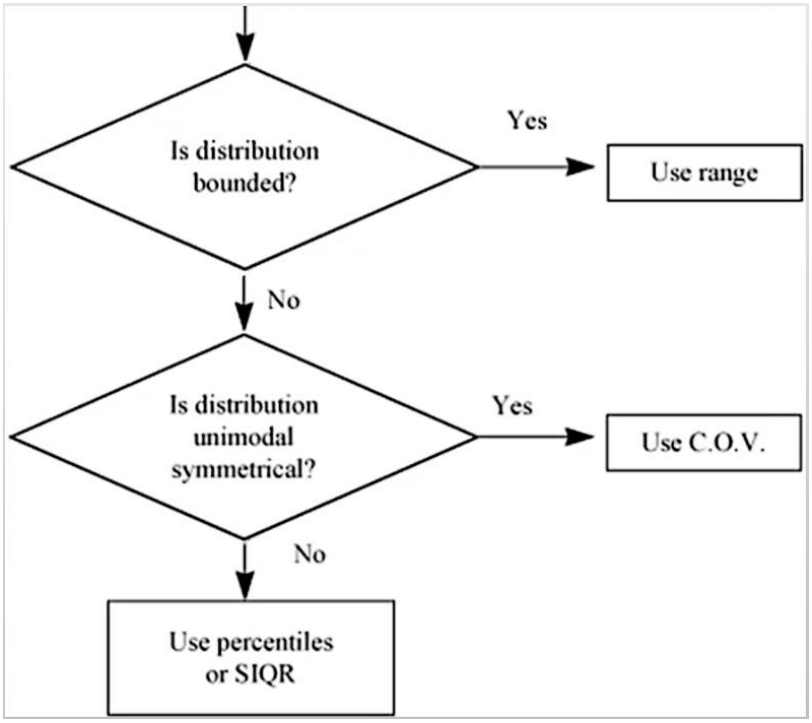
- 5-percentile and 95-percentile of a variable
    - Similar impact as Min and Max
- Can be done for any variable (even variables without bounds)
- Quantile
    - Fraction between 0 and 1
    - 0.9 quantile = 90-percentile
- Decile
    - Percentiles at multiples of 10%
- Quartile
    - Percentiles at multiples of 25%
- Semi Inter-Quartile Range

$$SIQR = \frac{Q_3 - Q_1}{2} = \frac{x_{0.75} - x_{0.25}}{2}$$

Comparation of variation measures



Selecting Index of Dispersion



- Just guidelines
- E.g.
    - Networks
        - Upper range - worse case
        - Designed to carry 95 to 99-percentile of the observed load levels
    - Computing power
    - Power supplies - sustain peak demand

Summary

- Indices of central tendencies
    - Mode
    - Median

    - Mean
        - Arithmetic, Geometric, Harmonic means
- Indices of dispersion
    - Range
    - Variance
    - Percentiles
    - Quartiles
    - SIQR

EP Curs6 – Comparing Systems

Comparing computer systems using sample data

Overview

- Sample vs Population
- Confidence interval for the mean
- Approximate visual test
- One sided confidence intervals
- Sample size for determining mean and proportions

Sample vs Population

- Generate several million random numbers with mean μ and standard deviation σ
  Select a sample of n observations

$$\overline{x} \neq \mu$$

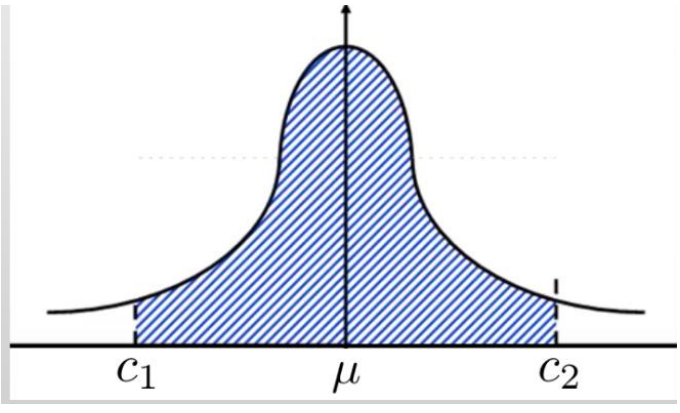
- Sample mean <> Population mean
- Population (unknown) - greek letters
  - Population characteristics —> parameters
- Sample - english letters
  - Sample estimates —> statistics

Confidence interval for the mean

- k samples => k sample means
  - Cannot get a single estimate of μ
  - Use bounds c1 and c2

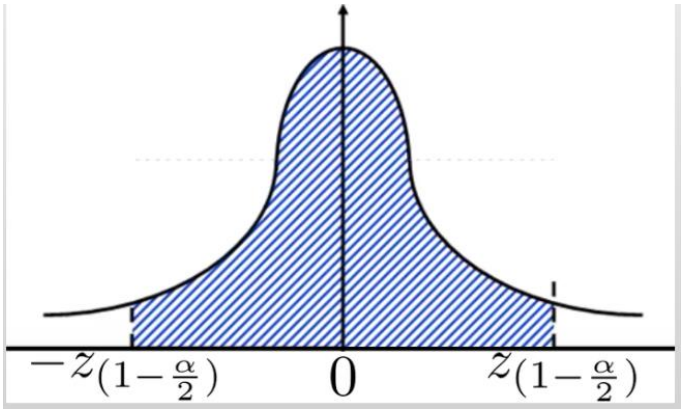
$$Probability\{c_1 \leq \mu \leq c_2\} = 1 - \alpha$$

- Confidence interval: (C1, C2)
- Significance level: α
- Confidence level: 100(1-α)
- Confidence coefficient: 1



Confidence interval

- 90% confidence interval
  - 5-percentile and 95-percentile of the sample means
- Central limit theorem
  - Sample mean is normally distributed for large sample sizes, regardless of the distribution from which we are sampling.



$$\overline{x} \sim N(\mu, \sigma/\sqrt{n})$$

Where μ = population mean and σ = population standard deviation

- Standard error: Standard deviation of the sample mean $s = \sigma/\sqrt{n}$
- 100(1-α)% confidence interval for μ:

$$(\overline{x} - z_{1-\frac{\alpha}{2}}\frac{s}{\sqrt{n}}, \overline{x} + z_{1-\frac{\alpha}{2}}\frac{s}{\sqrt{n}})$$

$$z_{1-\frac{\alpha}{2}} = (1-\frac{\alpha}{2}) - quantile\ \ of\ \ N(0,1)$$

Example

$$\overline{x} = 3.90,\ \ s = 0.95,\ \ n = 32$$

- 90% confidence interval for the mean

$$(\overline{x} - z_{1-\frac{\alpha}{2}}\frac{s}{\sqrt{n}}, \overline{x} + z_{1-\frac{\alpha}{2}}\frac{s}{\sqrt{n}})$$

$$3.90 \mp (1.645)(0.95)/\sqrt{32} = (3.62, 4.17)$$

- 95% confidence interval for the mean

$$3.90 \mp (1.960)(0.95)/\sqrt{32} = (3.57, 4.23)$$

- 99% confidence interval for the mean

$$3.90 \mp (2.576)(0.95)/\sqrt{32} = (3.46, 4.33)$$

TABLE A.2 Quantiles of the Unit Normal Distribution

p	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.5	0.000	0.025	0.050	0.075	0.100	0.126	0.151	0.176	0.202	0.228
0.6	0.253	0.279	0.305	0.332	0.358	0.385	0.412	0.440	0.468	0.496
0.7	0.524	0.553	0.583	0.613	0.643	0.674	0.706	0.739	0.772	0.806
0.8	0.842	0.878	0.915	0.954	0.994	1.036	1.080	1.126	1.175	1.227

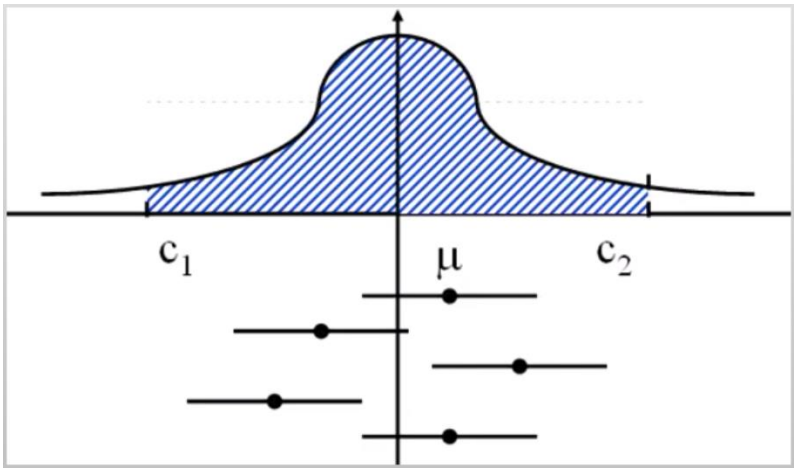
p	0.000	0.001	0.002	0.003	0.004	0.005	0.006	0.007	0.008	0.009
0.90	1.282	1.287	1.293	1.299	1.305	1.311	1.317	1.323	1.329	1.335
0.91	1.341	1.347	1.353	1.359	1.366	1.372	1.379	1.385	1.392	1.398
0.92	1.405	1.412	1.419	1.426	1.433	1.440	1.447	1.454	1.461	1.468
0.93	1.476	1.483	1.491	1.499	1.506	1.514	1.522	1.530	1.538	1.546
0.94	1.555	1.563	1.572	1.580	1.589	1.598	1.607	1.616	1.626	1.635
0.95	1.645	1.655	1.665	1.675	1.685	1.695	1.706	1.717	1.728	1.739
0.96	1.751	1.762	1.774	1.787	1.799	1.812	1.825	1.838	1.852	1.866
0.97	1.881	1.896	1.911	1.927	1.943	1.960	1.977	1.995	2.014	2.034
0.98	2.054	2.075	2.097	2.120	2.144	2.170	2.197	2.226	2.257	2.290

p	0.0000	0.0001	0.0002	0.0003	0.0004	0.0005	0.0006	0.0007	0.0008	0.0009
0.990	2.326	2.330	2.334	2.338	2.342	2.346	2.349	2.353	2.357	2.362
0.991	2.366	2.370	2.374	2.378	2.382	2.387	2.391	2.395	2.400	2.404
0.992	2.409	2.414	2.418	2.423	2.428	2.432	2.437	2.442	2.447	2.452
0.993	2.457	2.462	2.468	2.473	2.478	2.484	2.489	2.495	2.501	2.506
0.994	2.512	2.518	2.524	2.530	2.536	2.543	2.549	2.556	2.562	2.569
0.995	2.576	2.583	2.590	2.597	2.605	2.612	2.620	2.628	2.636	2.644
0.996	2.652	2.661	2.669	2.678	2.687	2.697	2.706	2.716	2.727	2.737
0.997	2.748	2.759	2.770	2.782	2.794	2.807	2.820	2.834	2.848	2.863
0.998	2.878	2.894	2.911	2.929	2.948	2.968	2.989	3.011	3.036	3.062
0.999	3.090	3.121	3.156	3.195	3.239	3.291	3.353	3.432	3.540	3.719

(asta e pt 90%, se ia procentul droit cu radical si se ia din table nr)

Confidence interval

- Take 100 samples and construct confidence interval for each sample, the interval would include the population mean in 90 cases.



Confidence interval for small samples

- How CI works - assume large number of observations => mean is normally distributed
- Large sample = 30
    - Not absolute bound
    - Exceptions exist (30+ observations and nowhere close to a normal distribution)
- What can you do if you do not have 30 observations?
- 100(1-α)% confidence interval for n<30

$$(\overline{x} - t_{[1-\frac{\alpha}{2};n-1]}\frac{s}{\sqrt{n}}, \overline{x} + t_{[1-\frac{\alpha}{2};n-1]}\frac{s}{\sqrt{n}})$$

- $t_{[1-\frac{\alpha}{2};n-1]} = (1-\frac{\alpha}{2})$ - quantile of a t-variable with n-1 degrees of freedom

Example

- Sample: -0.04, -0.19, 0.14, -0.09, -0.14, 0.19, 0.04, 0.09
- Mean = 0
- Sample standard deviation = 0.138
- For 90% interval: $t_{[1-\frac{\alpha}{2};n-1]} = t_{[0.95;7]} =$

$$(\overline{x} - t_{[1-\frac{\alpha}{2};n-1]}\frac{s}{\sqrt{n}}, \overline{x} + t_{[1-\frac{\alpha}{2};n-1]}\frac{s}{\sqrt{n}})$$

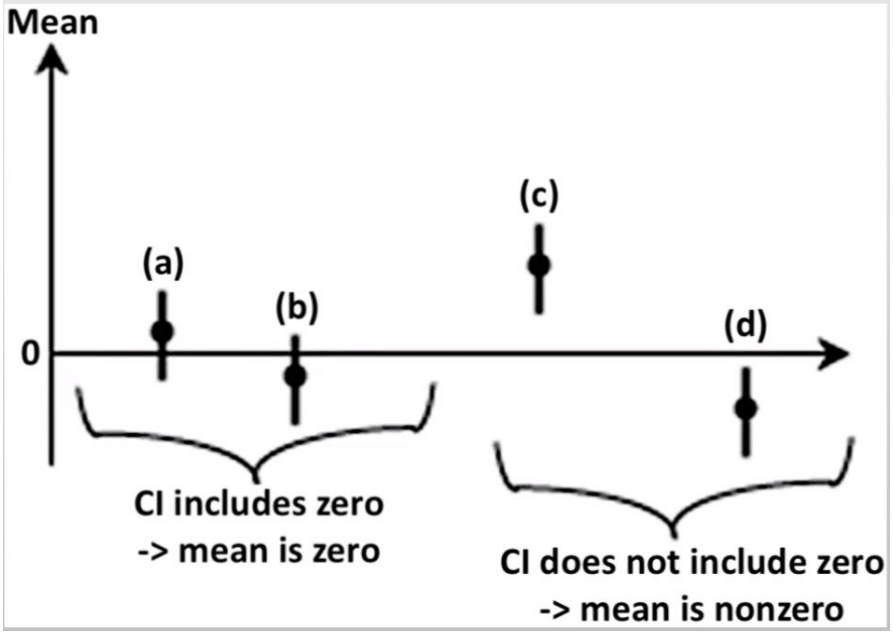
- Confidence interval for the mean:

$$0 \mp 1.895 \times 0.138/\sqrt{8} = 0 \mp 0.092 = (-0.092, 0.092)$$

TABLE A.4 Quantiles of the *t* Distribution

	p							
n	**0.6000**	**0.7000**	**0.8000**	**0.9000**	**0.9500**	**0.9750**	**0.9950**	**0.9995**
1	0.325	0.727	1.377	3.078	6.314	12.706	63.657	636.619
2	0.289	0.617	1.061	1.886	2.920	4.303	9.925	31.599
3	0.277	0.584	0.978	1.638	2.353	3.182	5.841	12.924
4	0.271	0.569	0.941	1.533	2.132	2.776	4.604	8.610
5	0.267	0.559	0.920	1.476	2.015	2.571	4.032	6.869
6	0.265	0.553	0.906	1.440	1.943	2.447	3.707	5.959
7	0.263	0.549	0.896	1.415	1.895	2.365	3.499	5.408
8	0.262	0.546	0.889	1.397	1.860	2.306	3.355	5.041
9	0.261	0.543	0.883	1.383	1.833	2.262	3.250	4.781
10	0.260	0.542	0.879	1.372	1.812	2.228	3.169	4.587
11	0.260	0.540	0.876	1.363	1.796	2.201	3.106	4.437
12	0.259	0.539	0.873	1.356	1.782	2.179	3.055	4.318
13	0.259	0.538	0.870	1.350	1.771	2.160	3.012	4.221
14	0.258	0.537	0.868	1.345	1.761	2.145	2.977	4.140
15	0.258	0.536	0.866	1.341	1.753	2.131	2.947	4.073
16	0.258	0.535	0.865	1.337	1.746	2.120	2.921	4.015
17	0.257	0.534	0.863	1.333	1.740	2.110	2.898	3.965
18	0.257	0.534	0.862	1.330	1.734	2.101	2.878	3.922
19	0.257	0.533	0.861	1.328	1.729	2.093	2.861	3.883
20	0.257	0.533	0.860	1.325	1.725	2.086	2.845	3.850
21	0.257	0.532	0.859	1.323	1.721	2.080	2.831	3.819
22	0.256	0.532	0.858	1.321	1.717	2.074	2.819	3.792
23	0.256	0.532	0.858	1.319	1.714	2.069	2.807	3.768
24	0.256	0.531	0.857	1.318	1.711	2.064	2.797	3.745
25	0.256	0.531	0.856	1.316	1.708	2.060	2.787	3.725
26	0.256	0.531	0.856	1.315	1.706	2.056	2.779	3.707
27	0.256	0.531	0.855	1.314	1.703	2.052	2.771	3.690
28	0.256	0.530	0.855	1.313	1.701	2.048	2.763	3.674
29	0.256	0.530	0.854	1.311	1.699	2.045	2.756	3.659
30	0.256	0.530	0.854	1.310	1.697	2.042	2.750	3.646
60	0.254	0.527	0.848	1.296	1.671	2.000	2.660	3.460
90	0.254	0.526	0.846	1.291	1.662	1.987	2.632	3.402
120	0.254	0.526	0.845	1.289	1.658	1.980	2.617	3.373

Testing for zero mean



Example

- Difference in processor times: {1.5, 2.6, -1.8,1.3, -0.5, 1.7, 2.4}
- Can we say with 99% confidence that one is superior to the other?
  Mean = 7.2 / 7 = 1.03
  Sample size = n = 7
  Sample variance = (22.84 - 7.2 * 7.2 / 7) / 6 = 2.57
  Sample standard deviation = sqr(2.57) = 1.6
- Confidence interval:

$$1.03 \mp t * 1.6/\sqrt{7} = 1.03 \mp 0.6t$$
$$100(1-\alpha) = 99,\ \ \alpha = 0.01,\ \ 1-\alpha/2 = 0.995,\ \ t_{[0.995,6]} = 3.707$$

TABLE A.4 Quantiles of the *t* Distribution

	p							
n	**0.6000**	**0.7000**	**0.8000**	**0.9000**	**0.9500**	**0.9750**	**0.9950**	**0.9995**
1	0.325	0.727	1.377	3.078	6.314	12.706	63.657	636.619
2	0.289	0.617	1.061	1.886	2.920	4.303	9.925	31.599
3	0.277	0.584	0.978	1.638	2.353	3.182	5.841	12.924
4	0.271	0.569	0.941	1.533	2.132	2.776	4.604	8.610
5	0.267	0.559	0.920	1.476	2.015	2.571	4.032	6.869
6	0.265	0.553	0.906	1.440	1.943	2.447	3.707	5.959
7	0.263	0.549	0.896	1.415	1.895	2.365	3.499	5.408
8	0.262	0.546	0.889	1.397	1.860	2.306	3.355	5.041
9	0.261	0.543	0.883	1.383	1.833	2.262	3.250	4.781
10	0.260	0.542	0.879	1.372	1.812	2.228	3.169	4.587
11	0.260	0.540	0.876	1.363	1.796	2.201	3.106	4.437
12	0.259	0.539	0.873	1.356	1.782	2.179	3.055	4.318
13	0.259	0.538	0.870	1.350	1.771	2.160	3.012	4.221
14	0.258	0.537	0.868	1.345	1.761	2.145	2.977	4.140
15	0.258	0.536	0.866	1.341	1.753	2.131	2.947	4.073
16	0.258	0.535	0.865	1.337	1.746	2.120	2.921	4.015
17	0.257	0.534	0.863	1.333	1.740	2.110	2.898	3.965
18	0.257	0.534	0.862	1.330	1.734	2.101	2.878	3.922
19	0.257	0.533	0.861	1.328	1.729	2.093	2.861	3.883
20	0.257	0.533	0.860	1.325	1.725	2.086	2.845	3.850
21	0.257	0.532	0.859	1.323	1.721	2.080	2.831	3.819
22	0.256	0.532	0.858	1.321	1.717	2.074	2.819	3.792
23	0.256	0.532	0.858	1.319	1.714	2.069	2.807	3.768
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120	0.254	0.526	0.845	1.289	1.658	1.980	2.617	3.373

- 99% confidence interval = (-1.21, 3.27)
- Opposite signs => we cannot say with 99% confidence that the mean difference is significantly different from zero.
- Answer: They are the same.
- Answer: The difference is zero.
- Difference in processor times: {1.5, 2.6, -1.8, 1.3, -0.5, 1.7, 2.4}

- 99% confidence interval = (-1.21, 3.27)
- Is the difference 1?
- Answer: Yes, the difference is 1.

Paired vs unpaired comparisons

- Paired
  - One-to-one correspondence between the ith test on system A and ith test on system B
  - Example: performance on i-th workload
  - Use confidence interval of the difference
- Unpaired
  - No correspondence
  - Number of observations may differ
  - Example
    - n people on system A, n on system B
    - Need more sophisticated method

Example

- Performance: {(5.4, 19.1), (16.6, 3.5), (0.6, 3.4), (1.4, 2.5), (0.6, 3.6) (7.3, 1.7)}
  Is one system better?
- Differences: {-13.7, 13.1, -2.8, -1.1, -3.0, 5.6}
  Sample mean = -0.32
  Sample variance = 81.62
  Sample standard deviation = 9.03
  Confidence interval for the mean:

$$-0.32 \mp t\sqrt{81.62/6} = -0.32 \mp t(3.69), \;\; t_{[0.95,5]} = 2.015$$

$$\text{90\% confidence interval:} \quad -0.32 \mp (2.015)(3.69) = (-7.75, 7.11)$$

- Answer: No, they are not different.

Unpaired observations:

- Compute sample means

$$\overline{x}_a = \frac{1}{n_a}\sum_{i=1}^{n_a} x_{ia}$$

$$\overline{x}_b = \frac{1}{n_b}\sum_{i=1}^{n_b} x_{ib}$$

- Compute the sample standard deviations

$$s_a = \left\{ \frac{(\sum_{i=1}^{n_a} x_{ia}^2) - n_a x_a^{-2}}{n_a - 1} \right\}^{\frac{1}{2}}$$

$$s_b = \left\{ \frac{(\sum_{i=1}^{n_b} x_{ib}^2) - n_b x_b^{-2}}{n_b - 1} \right\}^{\frac{1}{2}}$$

- Compute mean differences

$$(\overline{x}_a - \overline{x}_b)$$

- Compute standard deviation of the mean differences

$$s = \sqrt{\left(\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b} \right)}$$

- Compute the effective number of degrees of freedom

$$\nu = \frac{\left(\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b} \right)^2}{\frac{1}{n_a+1} \left(\frac{s_a^2}{n_a} \right)^2 + \frac{1}{n_b+1} \left(\frac{s_b^2}{n_b} \right)^2} - 2$$

- Compute the confidence interval for the mean differences

$$(\overline{x}_a - \overline{x}_b) \mp t_{[1-\alpha/2;\nu]} s$$

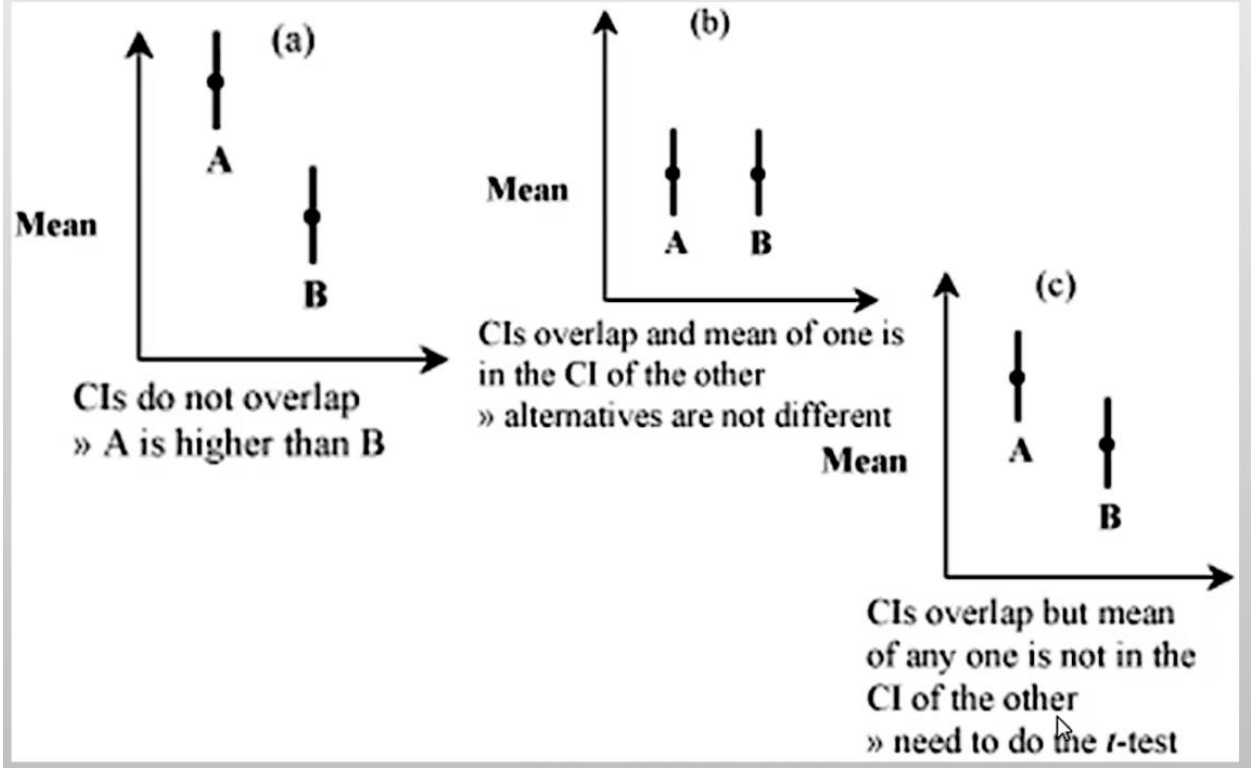
Example

- Times on system A: {5.36, 16.57, 0.62, 1.41, 0.64, 7.26}
  Times on system B: {19.12, 3.52, 3.38, 2.50, 3.60, 1.74}
- Are the two systems significantly different?

- System A
  - Mean $\overline{x}_a = 5.31$
  - Variance $s_a^2 = 37.92$
  - $n_a = 6$
- System B
  - Mean $\overline{x}_b = 5.64$
  - Variance $s_b^2 = 44.11$
  - $n_b = 6$

- Mean difference xa — xb = -0.33
- Standard deviation of the mean difference = 3.698
- Effective number of degrees of freedom *f* = 11.921
- 0.95-quantile of a t-variate with 12 degrees of freedom = 1.71
- 90% confidence interval for the difference = (-6.92, 6.26)
- The confidence interval includes zero => the two systems are not different

Approximate visual test



Example

- Times on system A: {5.36, 16.57, 0.62, 1.41, 0.64, 7.26}
  Times on system B: {19.12, 3.52, 3.38, 2.50, 3.60, 1.74}
  $$t_{[0.95,5]} = 2.015$$
- 90% confidence interval for the mean of A
  $$5.31 \mp (2.015)\sqrt{37.92/6} = (0.24, 10.38)$$
- 90% confidence interval for the mean of B
  $$5.64 \mp (2.015)\sqrt{44.11/6} = (0.18, 11.10)$$

- Confidence intervals overlap and the mean of one falls in the confidence interval of the other =>
- The two systems are not different at this level of confidence

What level of confidence to use?

- 90%, 95%, 99% ?
- Base on the loss that you will sustain if the parameter is outside the range and the gain you would have if the parameter is inside the range.
- Low loss -> low confidence is fine
    - Example: Lottery of $5million with probability 10^(-7)
        - 90% confidence —> buy 9 million tickets
        - 0.01% confidence level if fine
    - 50% confidence may or may not be too low
    - 99% confidence may or may not be too high

One-sided confidence intervals

- Two side intervals - 90% confidence
    - P(difference > upper limit) = 5%
    - P(difference < lower limit) = 5%
- One-sided confidence interval
    - One sided question: Is the mean greater than 0?
- One-sided lower confidence interval for μ

$$\left(\overline{x} - t_{[1-\alpha;n-1]}\frac{s}{\sqrt{n}}, \overline{x}\right)$$

Note t at 1-α (not at 1-α/2)

- One-sided upper confidence interval for μ

$$\left(\overline{x}, \overline{x} + t_{[1-\alpha;n-1]}\frac{s}{\sqrt{n}}\right)$$

- For large samples - use z instead of t

Example

- Time between crashes

System	Number	Mean	Stdv
A	972	124.10	198.20
B	153	141.47	226.11

- Assume unpaired observations
- Mean difference

$$\overline{x}_A - \overline{x}_B = 124.10 - 141.47 = -17.37$$

- Standard deviation of the difference

$$s = \sqrt{\left(\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b}\right)} = \sqrt{\left(\frac{(198.20)^2}{972} + \frac{(226.11)^2}{153}\right)} = 19.35$$

- Effective number of degrees of freedom

$$\nu = \frac{\left(\frac{s_a^2}{n_a} + \frac{s_b^2}{n_b}\right)^2}{\frac{1}{n_a+1}\left(\frac{s_a^2}{n_a}\right)^2 + \frac{1}{n_b+1}\left(\frac{s_b^2}{n_b}\right)^2} - 2 = \ldots = 191.05$$

- v > 30 use z instead of t
- One-sided test => Use $z_{0.90}$ = 1.28 for 90% confidence
- 90% confidence interval: (-17.37, -17.37+1.28*19.35) = (-17.37, 7.402)
- CI includes zero => system A is not more susceptible to crashes than system B.

Sample size for determining mean

- Large sample => narrower confidence interval. - Higher confidence
- Question
  - How many observations n to get an accuracy of +/- r% and a confidence level of 100(1-α)%?

$$\overline{x} \mp z\frac{s}{\sqrt{n}}$$

- r% accuracy

$$CI = (\overline{x}(1 - r/100), \overline{x}(1 + r/100))$$

$$\overline{x} \mp z\frac{s}{\sqrt{n}} = \overline{x}\left(1 \mp \frac{r}{100}\right) \Rightarrow z\frac{s}{\sqrt{n}} = \overline{x}\frac{r}{100} \Rightarrow n = \left(\frac{100zs}{r\overline{x}}^2\right)$$

Example

- Sample mean of the response time = 20 sec
- Sample standard deviation = 5
- Question: How many repetitions are needed to get the response time accurate within 1 second at 95% confidence?
- Required accuracy is 1 in 20 = 5%

$$\overline{x} = 20, \quad s = 5, \quad z = 1.960, \quad r = 5$$

$$n = \left(\frac{100 \times 1.960 \times 5}{5 \times 20}\right)^2 = 9.8^2 = 96.04$$

- Answer: 97 observations are needed

Takeaways

- All statistics based on a sample are random and should be specified with a confidence interval
- If the confidence interval includes zero, the hypothesis that the population mean is zero cannot be rejected
- Paired observations => test the difference for zero mean
- Unpaired observations => more sophisticated

Curs 10 – Regression Models

Simple Linear Regression Models

Overview

1. Definition of a good model
2. Estimation of model parameters
3. Allocation of variation
4. Standard deviation of errors
5. Confidence intervals for regression parameters
6. Confidence intervals for predictions
7. Visual tests for verifying regression assumption

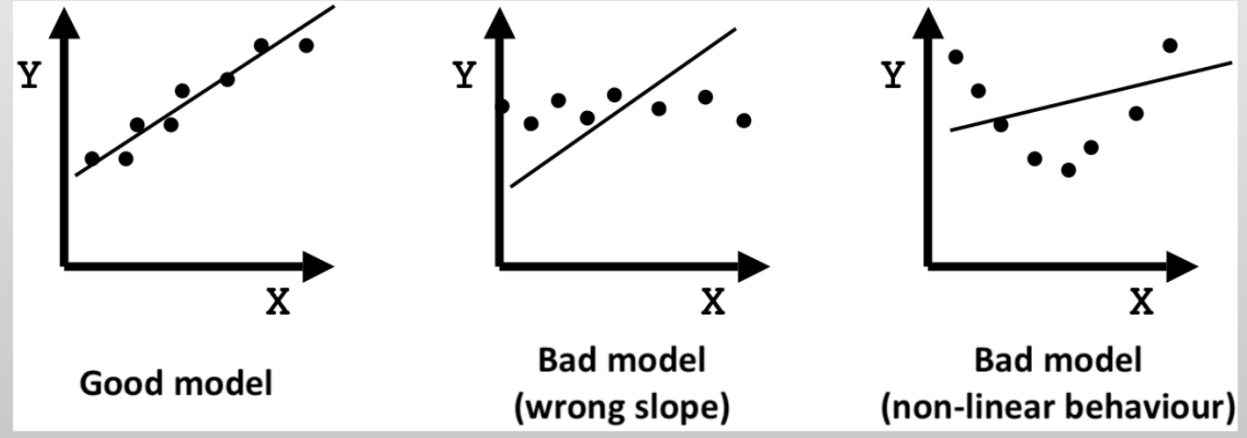
Simple linear regression models

- Regression model
    - Predict a response for a given set of predictor variables
    - Response variable
        - Estimated variable
    - Predictor variables
        - Variables used to predict the response predictors or factors
- Linear regression models
    - Response is a linear function of predictors
- Simple linear regression models
    - Only one predictor

Simple linear regression models

- Regression model
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    - Response is a linear function of predictors
- Simple linear regression models
    - Only one predictor

Good Model



- Regression models attempt to minimize the distance measured vertically between the observation point and the model line
- The length of the line segment is called residual / modelling error / error
- The negative and positive errors should cancel out
  - => zero overall error
  - Many lines satisfy this criterion
- Choose the line that minimizes the sum of squares of the errors

$$\widehat{y} = b_0 + b_1 x$$

  - $\hat{y}$ is the predicted response when the predictor variable is *x*
  - b0 and b1 are fixed regression parameters to be determined from the data
- Given n observation pairs { (x1, y1), …, (xn, yn)}, the estimated response $\hat{y}_i$ for the ith observation is

$$\widehat{y_i} = b_0 + b_1 x_i$$

- The error is

$$e_i = y_i - \widehat{y_i}$$

- The best linear model minimizes the sum of squared errors (SSE)

$$\sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left(y_i - b_0 - b_1 x_i\right)^2$$

- Subject to the constraint that the mean error is zero

$$\sum_{i=1}^{n} e_i = \sum_{i=1}^{n} \left(y_i - b_0 - b_1 x_i\right) = 0$$

- This is equivalent to minimizing the variance of errors

Estimation of mode parameters

- Regression parameters that give minimum error variance

$$b_1 = \frac{\sum xy - n\overline{x}\,\overline{y}}{\sum x^2 - n\overline{x}^2} \qquad \text{and} \qquad b_0 = \overline{y} - b_1\overline{x}$$

- Where

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \overline{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

$$\sum xy = \sum_{i=1}^{n} x_i y_i \qquad \sum x^2 = \sum_{i=1}^{n} x_i^2$$

Example

- The number of disk I/O's and processor times of seven programs were measured as {(14, 2), (16, 5), (27, 7 (42, 9), (39, 10), (50, 13), (83, 20)}.
- For this data

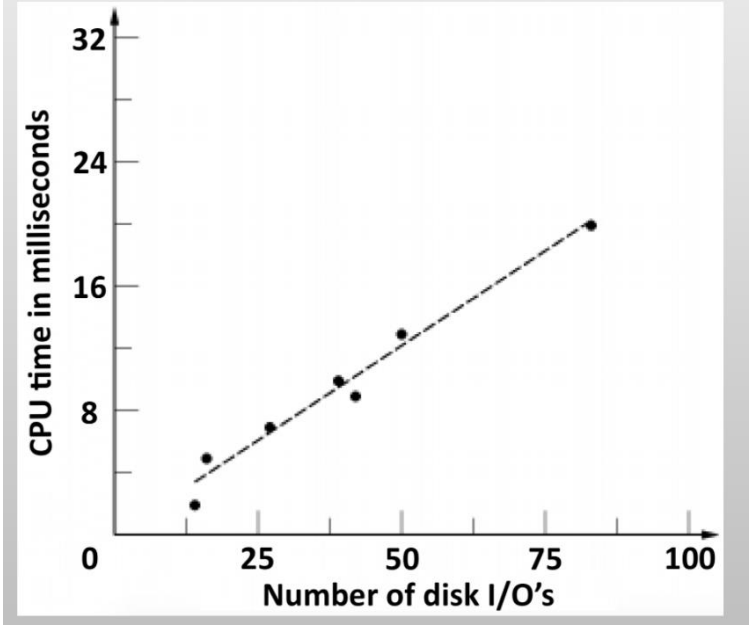
$$n = 7,\ \sum xy = 3375,\ \sum x = 271,\ \sum x^2 = 13.855$$
$$\sum y = 66,\ \sum y^2 = 828,\ \overline{x} = 38.71,\ \overline{y} = 9.43$$

$$b_1 = \frac{\sum xy - n\overline{x}\,\overline{y}}{\sum x^2 - n(\overline{x})^2} = \frac{3375 - 7 \times 38.71 \times 9.43}{13.855 - 7 \times (38.71)^2} = 0.2438$$

$$b_0 = \overline{y} - b_1\overline{x} = 9.43 - 0.2438 \times 38.71 = -0.0083$$

- The desired linear model is:

$$CPU\,time = -0.0083 + 0.2438(Number\,of\,disk\,I/O's)$$



	Disk I/O's	CPU Time	Estimate	Error	Error^2
	x_i	y_i	$\widehat{y}_i = b_0 + b_1 x_i$	$e_i = y_i - \widehat{y}_i$	e_i^2
	14	2	3.4043	-1.4043	1.9721
	16	5	3.8918	1.1082	1.2281
	27	7	6.5731	0.4269	0.1822
	42	9	10.2295	-1.2295	1.5116
	39	10	9.4982	0.5018	0.2518
	50	13	12.1795	0.8205	0.6732
	83	20	20.2235	-0.2235	0.0500
$\sum$	272	66	66.0000	0.00	5.8690

Derivation of regression parameters

- The error of the ith observation

$$e_i = y_i - \widehat{y}_i = y_i - (b_0 + b_1 x_i)$$

- The mean error for a sample of n observations

$$\overline{e} = \frac{1}{n}\sum_{i=1}^{n} e_i = \frac{1}{n}\sum_{i=1}^{n} \{y_i - (b_0 + b_1 x_i)\} = \overline{y} - b_0 - b_1 \overline{x}$$

- Setting mean error to zero

$$b_0 = \overline{y} - b_1\overline{x}$$

- Substitute b0 in the error expression

$$e_i = y_i - \overline{y} + b_1\overline{x} - b_1x_i = \left(y_i - \overline{y}\right) - b_1\left(x_i - \overline{x}\right)$$

$$SSE = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \left\{(y_i - \overline{y})^2 + 2b_1(y_i - \overline{y})(x_i - \overline{x}) + b_1^2(x_i - \overline{x})^2\right\}$$

$$\frac{SSE}{n-1} = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \overline{y})^2 - 2b_1\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \overline{y})(x_i - \overline{x}) + b_1^2\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{x})^2$$

$$= s_y^2 - 2b_1s_{xy}^2 + b_1^2s_x^2$$

- Divided by n-1 to facilitate the last notation (n-1 is not related to the degrees of freedom

$$\frac{1}{n-1}\frac{d(SSE)}{db_1} = -2s_{xy}^2 + 2b_1s_x^2 = 0$$

$$b_1 = \frac{s_{xy}^2}{s_x^2} = \frac{\sum xy - n\overline{x}\,\overline{y}}{\sum x^2 - n(\overline{x})^2}$$

Allocation of variation

- Error variance without regression = variance of the response

$$Error = e_i = Observed\,response - Predicted\,response = y_i - \overline{y}$$

- Variance of errors without regression =

$$\frac{1}{n-1}\sum_{i=1}^{n} e_i^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \overline{y})^2 = Variance\,of\,y$$

- The sum of squared errors without regression

$$\sum_{i=1}^{n}(y_i - \overline{y})^2$$

- SST = total sum of squares
    - It is a measure of y's variability and is called variation of y

$$SST = \sum_{i=1}^{n}(y_i - \overline{y})^2 = \left(\sum_{i=1}^{n} y_i^2\right) - n\overline{y}^2 = SSY - SS0$$

- SSY is the sum of squares of y (or $\sum y^2$)
- SSO is the sum of squares of $\bar{y}$ and is equal to $n\bar{y}^2$

- The difference between SST and SSE is the sum of squares explained by the regression (SSR)
  SSR = SST - SSE
  SST = SSR + SSE
- Coefficient of determination (the fraction of the variation that is explained determines the goodness of the regression)

$$R^2 = \frac{SSR}{SST} = \frac{SST - SSE}{SST}$$

- The higher the value of R^2 the better the regression
    - $R^2 = 1$ => perfect fit
    - $R^2 = 0$ => no fit

$$Sample\,correlation(x, y) = R_{xy} = \frac{s_{xy}^2}{s_x s_y}$$

$$Coefficient\ of\ determination = \{Correlation\ coefficient(x, y)\}^2$$

- Shortcut formula for SSE

$$SSE = \sum y^2 - b_0 \sum y - b_1 \sum xy$$

Example

- Consider the disk I/O – CPU time data from the previous example

$$SSE = \sum y^2 - b_0 \sum y - b_1 \sum xy$$
$$= 828 + 0.0083 \times 66 - 0.2438 \times 3375 = 5.87$$
$$SST = SSY - SS0 = \sum y^2 - n(\overline{y})^2$$
$$= 828 - 7 \times (9.43)^2 = 205.71$$
$$SSR = SST - SSE = 205.71 - 5.87 = 199.84$$
$$R^2 = \frac{SSR}{SST} = \frac{199.84}{205.71} = 0.9715$$

- The regression explains 97% of the CPU time's variation

Standard deviation of errors

- Variance of errors
  - Divide sum of squared errors by its degrees of freedom
  - Errors are obtained after calculating 2 regression parameters => n-2 degrees of freedom

$$s_e = \sqrt{\frac{SSE}{n-2}}$$

- MSE (mean squared errors) = SSE/(n-2)
- S_e (standard deviation of errors) = sqrt(MSE)
- SSY - n degrees of freedom
  - Obtained from n independent observations without estimating any parameters
- SSO - 1 degree of freedom
  - Computer simply from $\bar{y}$
- SST - n-1 degrees of freedom
  - Only requires one parameter to be calculated from the data $\bar{y}$
- SSR (= SST – SSE) has the remaining 1 degree of freedom

$$SST = SSY - SS0 = SSR + SSE$$
$$(n-1) = n - 1 = 1 + (n-2)$$

- Degrees of freedom add up (similar to sums of squares)

Example

- Consider the disk I/O - CPU time data example
- Degrees of freedom of the sums

$$\begin{array}{lrcrcrcr} SS: & SST & = & SSY & - & SS0 & = & SSR & + SSE \\ & 205.71 & = & 828 & - & 622.29 & = & 199.84 & + 5.87 \\ DF: & 6 & = & 7 & - & 1 & = & 1 & + 5 \end{array}$$

- Mean squared error

$$MSE = \frac{SSE}{DF\ for\ errors} = \frac{5.87}{5} = 1.17$$

- Standard deviation of error

$$s_e = \sqrt{MSE} = \sqrt{1.17} = 1.08$$

CI for regression parameters

- Regression coefficients b0 and b1 are estimates from a single sample of size n
  - => random
  - => using another sample, the estimates may be different
- Consider β0 and β1 the true parameters of the population

$$y = \beta_0 + \beta_1 x$$

- Computed coefficients b0 and b1 are estimates of βo and β1

$$s_{b_0} = s_e \left[\frac{1}{n} + \frac{\overline{x}^2}{\sum x^2 - n\overline{x}^2}\right]^{\frac{1}{2}}$$

$$s_{b_1} = \frac{s_e}{\left[\sum x^2 - n\overline{x}^2\right]^{\frac{1}{2}}}$$

- The 100(1-α)% confidence intervals for b0 and b1 can be computed using the 1-α/2 quantile of a t variate with n-2 degrees of freedom t[1—α/2; n-2]
- The confidence intervals

$$b_0 \mp t s_{b_0}$$

$$b_1 \mp t s_{b_1}$$

- If a confidence interval includes zero => the regression parameter cannot be considered different from zero at the 100(1-α)% CI

Example

- Consider the I/O – CPU time data example

$$n = 7,\ \overline{x} = 38.71,\ \sum x^2 = 13.855,\ s_e = 1.0834$$

$$s_{b_0} = s_e\left([\frac{1}{n} + \frac{\overline{x}^2}{\sum x^2 - n\overline{x}^2}\right]^{\frac{1}{2}}$$

$$= 1.0834\left[\frac{1}{7} + \frac{(38.71)^2}{13.855 - 7 \times 38.71 \times 38.71}\right]^{\frac{1}{2}} = 0.8311$$

$$s_{b_1} = \frac{s_e}{\left[\sum x^2 - n\overline{x}^2\right]^{\frac{1}{2}}}$$

$$= \frac{1.0834}{[13.855 - 7 \times 38.71 \times 38.71]^{\frac{1}{2}}} = 0.0187$$

$$b_0 \mp ts_{b_0} \qquad b_1 \mp ts_{b_1} \qquad t_{[1-\alpha/2;\ n-2]} = t_{[0.95;\ 5]}$$

- 90% confidence interval for b0

$$-0.0083 \mp (2.015)(0.8311) = (-1.6830, 1.6663)$$

- CI includes zero - the hypothesis that this parameter is zero cannot be rejected at 0.10 significance level => b0 is essentially zero
- 90% confidence interval for b1

$$0.2438 \mp (2.015)(0.0187) = (0.2061, 0.2814)$$

- CI does not include zero - the slope b1 is significantly different from zero at this confidence level

Example

- Best linear models
  Time on UNIX = 0.017 (Data size in bytes) + 24
  Time on ARGUS = 0.034 (Data size in bytes) +30
- The regression explains 81% and 75% of the variation, respectively.
  Does ARGUS take longer time per byte than UNIX?
  Does ARGUS have a longer setup time than UNIX?

- Intervals for intercepts overlap, while those for slopes do not
- Setup times are not significantly different in the 2 systems, while the per byte times (slopes) are different

UNIX			
Parameter	**Mean**	**Std. Dev.**	**Confidence Interval**
b0	26.898	2.005	(23.2968, 30.4988)
b1	0.017	0.003	(0.0128, 0.0219)
ARGUS			
Parameter	**Mean**	**Std. Dev.**	**Confidence Interval**
b0	31.068	4.711	(22.6076, 39.5278)
b1	0.034	0.006	(0.0231, 0.0443)

CI for predictions

- Mean value of the predicted response

$$\hat{y}_p = b_0 + b_1 x_p$$

- Standard deviation of the mean of a future sample of m observations

$$s_{\hat{y}_{mp}} = s_e \left[\frac{1}{m} + \frac{1}{n} + \frac{(x_p - \overline{x})^2}{\sum x^2 - n\overline{x}^2} \right]^{\frac{1}{2}}$$

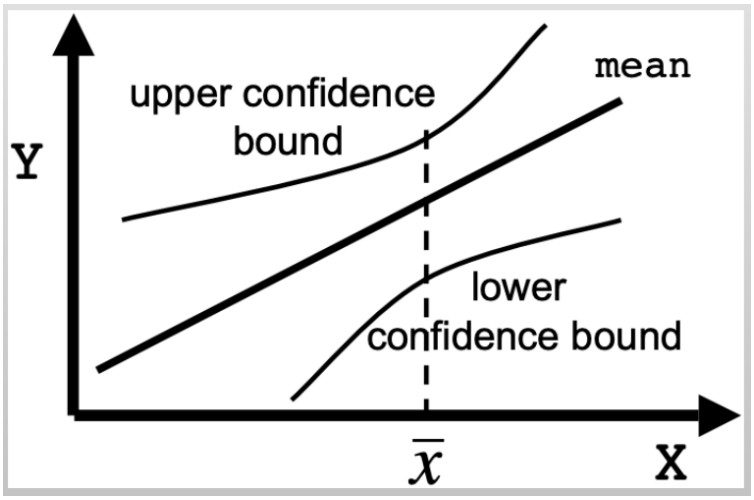
- m = 1 => standard deviation of a single observation

$$s_{\hat{y}_{1p}} = s_e \left[1 + \frac{1}{n} + \frac{(x_p - \overline{x})^2}{\sum x^2 - n\overline{x}^2} \right]^{\frac{1}{2}}$$

- m = ∞ => standard deviation of the mean of a large number of future observations at xp

$$s_{\hat{y}_p} = s_e \left[\frac{1}{n} + \frac{(x_p - \overline{x})^2}{\sum x^2 - n\overline{x}^2} \right]^{\frac{1}{2}}$$

- 100(1-α)% confidence interval for the mean can be constructed using a t quantile read at n-2 degrees of freedom
- Goodness of the prediction decreases as we move away from the center



Example

- Consider the disk I/O - CPU time data example
  CPU time = -0.0083 + 0.2438(Number of disk I/O's)
- What is the mean CPU time for a program with 100 disk I/O's?
  CPU time = -0.0083 + 0.2438(100) = 24.3674
- Standard deviation of errors S_e = 1.0834
- The standard deviation of the predicted mean of a large number of observations

$$s_{\widehat{y}_p} = s_e\left[\frac{1}{n} + \frac{(x_p - \overline{x})^2}{\sum x^2 - n\overline{x}^2}\right]^{\frac{1}{2}}$$

$$s_{\widehat{y}_p} = 1.0834\left[\frac{1}{7} + \frac{(100 - 38.71)^2}{13855 - 7(38.71)^2}\right]^{\frac{1}{2}} = 1.2159$$

- 90% CI for the predicted mean

$$24.3574 \mp (2.015)(1.2159) = (21.9174, 26.8174)$$

- CPU time of a single future program with 100 disk I/Os

$$s_{\widehat{y}_{1p}} = s_e\left[1 + \frac{1}{n} + \frac{(x_p - \overline{x})^2}{\sum x^2 - n\overline{x}^2}\right]^{\frac{1}{2}}$$

$$s_{\widehat{y}_p} = 1.0834\left[1 + \frac{1}{7} + \frac{(100 - 38.71)^2}{13855 - 7(38.71)^2}\right]^{\frac{1}{2}} = 1.6286$$

- 90% CI for a single prediction

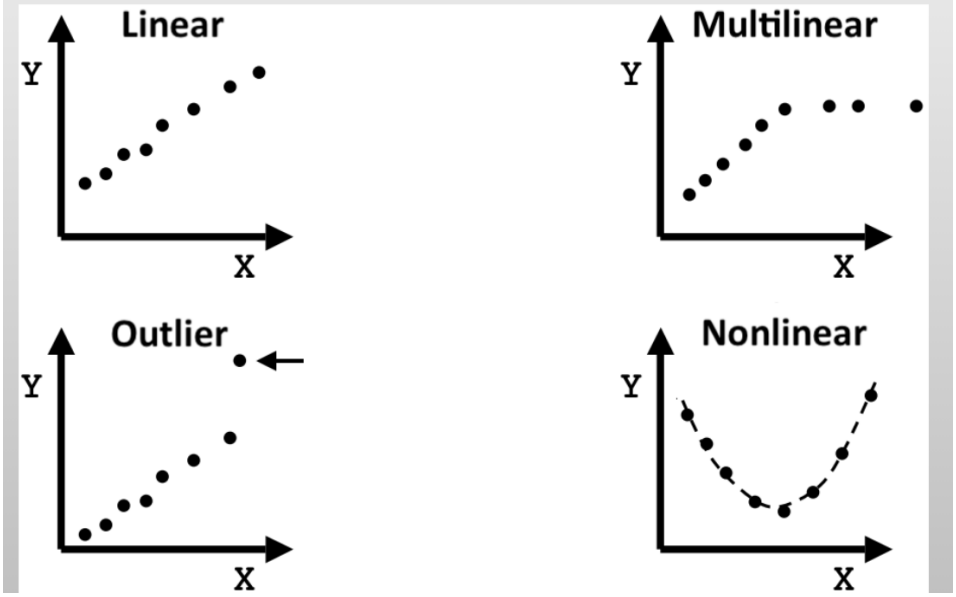
$$24.3574 \mp (2.015)(1.6286) = (21.0858, 27.6489)$$

Visual tests for regression assumptions

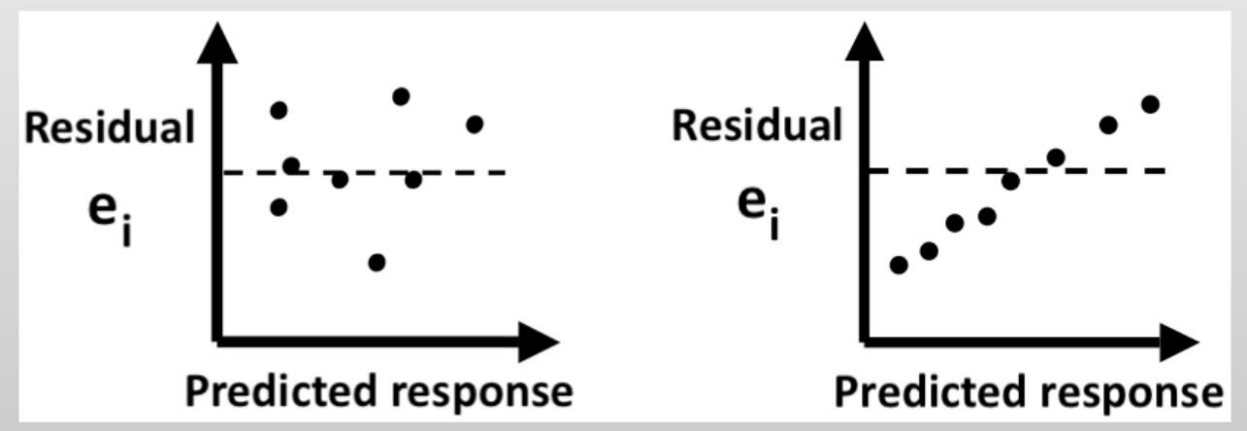
Regression assumptions

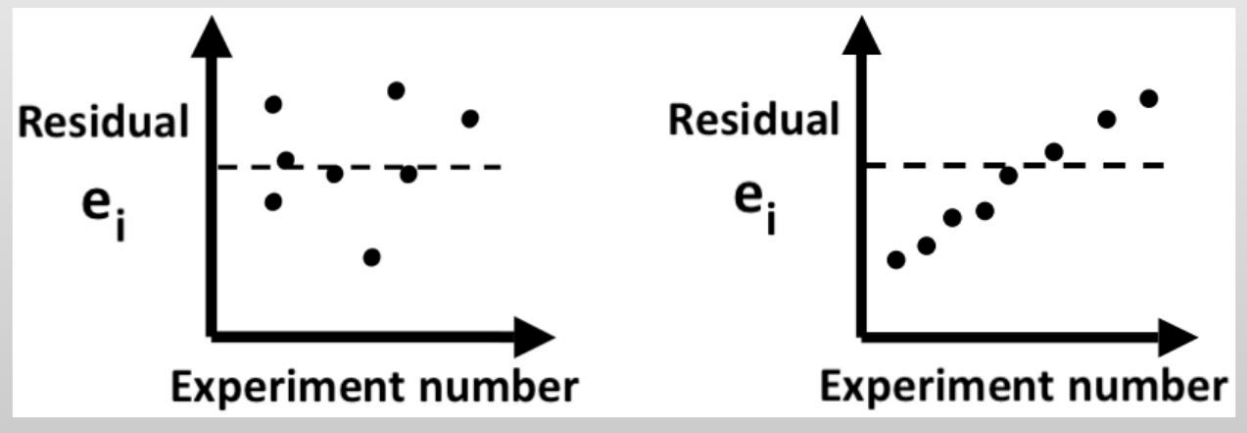
1. The true relationship between the response variable y and the predictor variable x is linear
2. The predictor variable x is non-stochastic and it is measured without any error
3. The model errors are statistically independent
4. The errors are normally distributed with zero mean and a constant standard deviation

1. Linear relationship



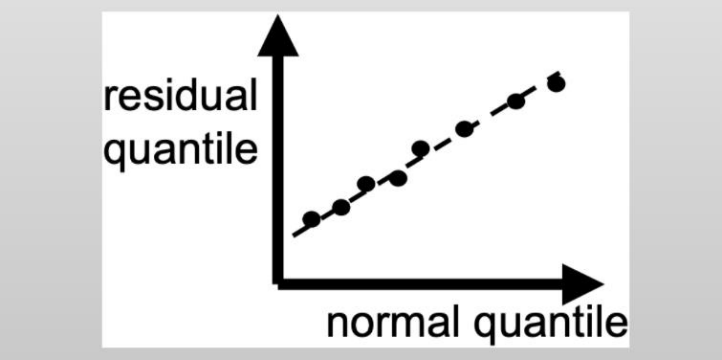
2. Independent errors



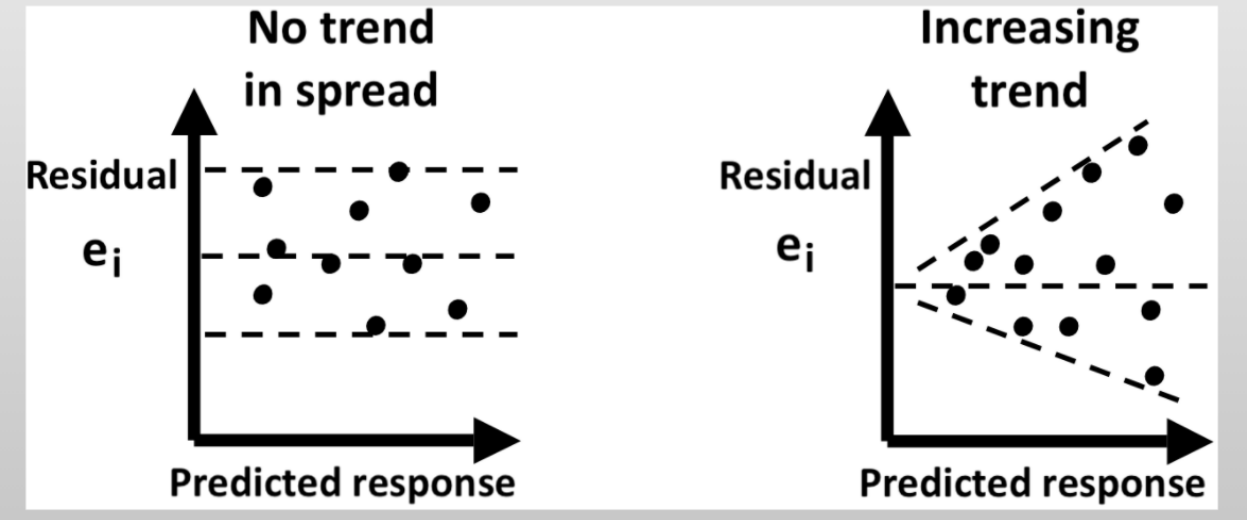


3. Normally distributed errors

- Use a quantile-quantile plot of e_i versus N(0,1)
    - Linear => the assumption is satisfied

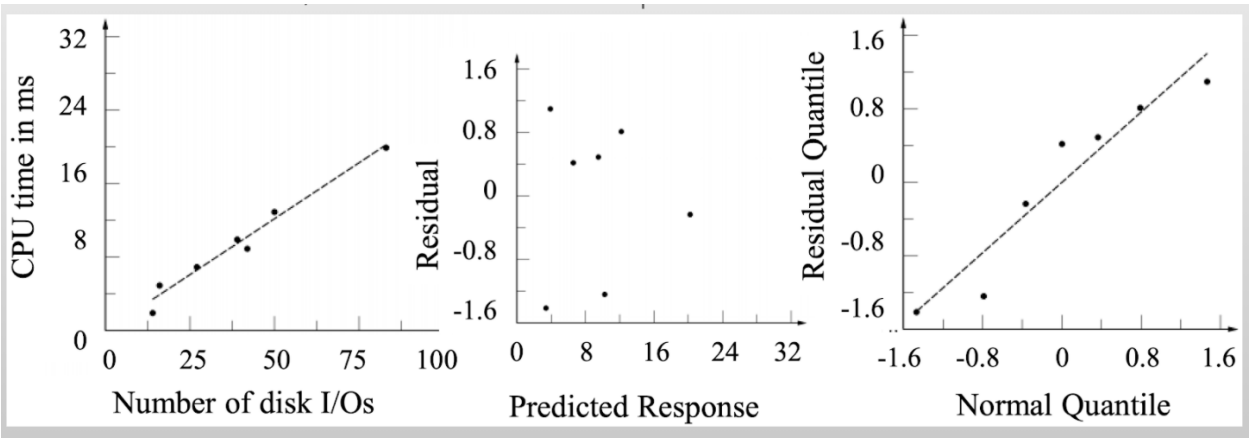


4.Constant standard deviation of errors



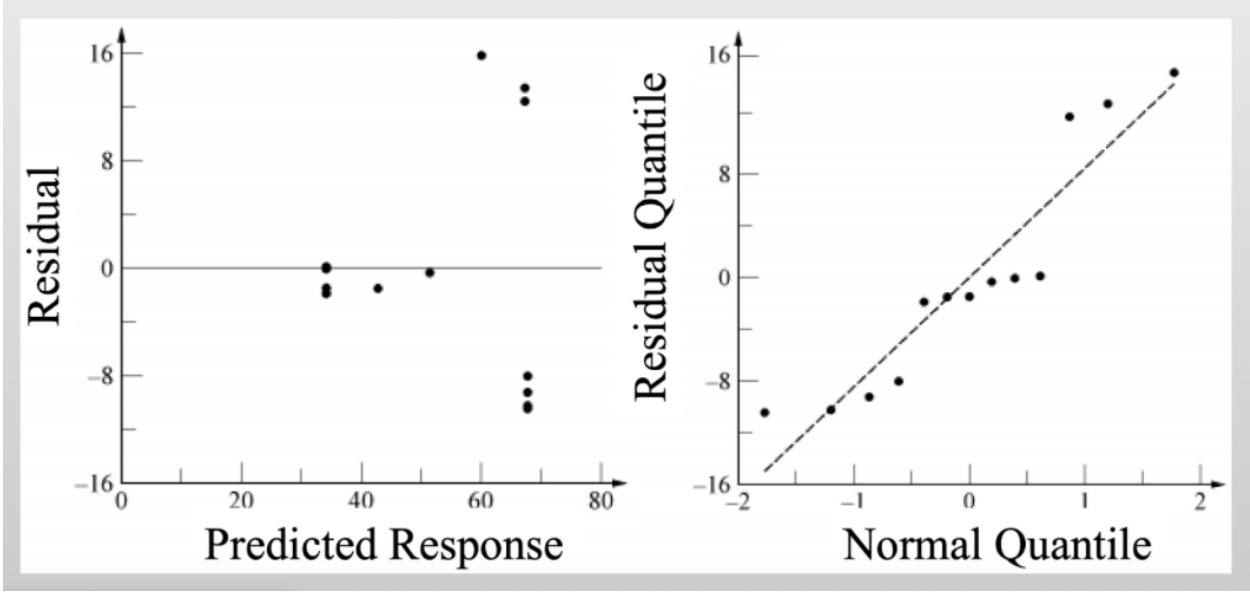
Example

Consider the disk I/O – CPU time data example



1. Relationship is linear
2. No trend in residuals seem independent
3. Linear normal quantile-quantile plot
    - Larger deviations at lower values but all values are small

Example



1. Large errors at larger responses
2. Normality of errors is questionable